Copernicus Global Land Operations - Lot I

"Vegetation and Energy"

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ALGORITHM THEORETICAL BASIS DOCUMENT

LAND SURFACE TEMPERATURE - LST V2.2

Issue I2.23

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07.03.2014		Move geoland2 document into Global Land document	I1.01
06.08.2014	All	Algorithm update after detection of artefacts: improve the atmospheric correction Inclusion of the second thermal infrared channel as input for the MTSAT LST. Included a new section with risk assessment and mitigation. Include clarifications required by review board.	I1.10
31.07.2015		Update LST algorithm to Himawari/AHI (delivery of WP 30.090.101). Align LST resolution to 5/112° (delivery of WP30.330.101).	I1.20
18.01.2016		Updated with clarifications after 2nd review of 2015.	I1.30
15.02.2019		Updated to include processing of GOES-16, leaving the necessary references to historical sensors / algorithms. General document update following more recent practices, framework and companion documents.	I1.40
27.06.2019	54-60	Editorial changes	I1.41
		Major algorithm update: Emissivity is now calculated dynamically using the Vegetation Cover Method with CGLOPS FCOVER as input. A common calibration database was introduced for both AHI/Himawari and ABI/GOES16, so model coefficients were updated for both sensors with respect to previous product versions. MSG IODC mission was introduced in the product, covering the Middle East. The timeslot for MSG 0º is now T=0 (whereas in previous versions the T-15 min was used).	I2.00

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		Added ANNEX 3 describing the cloud masking procedure.	
18.10.2020	All	Overall editorial changes complying with the 2020 review Added "Known Limitations" section, describing the effects of emissivity and snow Annex 3 was complemented with validation results of the cloud mask	I2.10
21.01.2022	All	Added changes documenting the introduction of the snow mask and its general impact to address known limitations of the LST product. Removed the corresponding reference to the issue from "known limitations" section.	I2.20
19.05.2022	32	Limit GOES-R pixels to cases where QFlag is "Good" to avoid outlier/unrealistic LSTs	I2.21
05.12.2022	All	Added angular correction for snow emissivity	I2.22
11.09.2025	28	Updated Table 5	I2.23

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List of Acronyms

AATSR	Advanced Along-Track Scanning Radiometer
ABI	Advanced Baseline Imager
AD	Applicable Document
AHI	Advanced Himawari Imager
ASTER	Advanced Spaceborne Thermal Emission and Reflection Radiometer
ATBD	Algorithm Theoretical Basis Document
AVHRR	Advanced Very High-Resolution Radiometer
CAL/VAL	Calibration and Validation
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
Cma	Cloud Mask
COG	Cloud Optimized Geotiff
DA	Dual-Algorithm
ECMWF	European Centre for Medium-Range Weather Forecasts
EUMETCast	EUMETSAT's broadcast system for environmental data
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
ESA	European Space Agency
FVC	Fraction of Vegetation Cover
FCOVER	CGLOPS Fraction of Vegetation Cover product
GBOV	Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GEO	Geostationary satellites
GOES	Geostationary Operational Environmental Satellite
GSW	Generalized Split-Window
HW/SW	Hardware / Software
IPMA	Instituto Português do Mar e da Atmosfera
IGBP	International Geosphere-Biosphere Program
INSPIRE	Infrastructure for Spatial Information in Europe
IODC	Indian Ocean Data Coverage
IR	InfraRed
ITCZ	Inter-Tropical Convergence Zone

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JAMI	Japanese Advanced Meteorological Imager
JHU	John Hopkins University
JMA	Japan Meteorological Agency
JRC	Joint Research Centre
LPV	Land Product Validation
LSA-SAF	EUMETSAT Satellite Application Facility on Land Surface Analysis
LST	Land Surface Temperature
LUT	Look-Up-Tables
MIR	Middle infrared
MODIS	Moderate Resolution Imaging Spectroradiometer
MODTRAN6	MODerate spectral resolution atmospheric TRANSsmittance
MSG	Meteosat Second Generation
MTG	Meteosat Third Generation
MTSAT	Multi-Function Transport Satellite
NetCDF	Network Common Data Form
NOAA	National Oceanic and Atmospheric Administration
Suomi NPP	Suomi National Polar-orbiting Partnership
NRT	Near Real Time
NWC SAF	Eumetsat SAF on Nowcasting and Very Short-Range Forecasting
NWP	Numerical Weather Prediction
OLCI	Ocean and Land Colour Instrument
PUM	Product User Manual
QAR	Quality Assessment Report
QC	Quality Control
RMSD	Root Mean Square Difference
SAF	Satellite Application Facility
SCE	Snow Cover Extent
SEVIRI	Spinning Enhanced Visible Infra-Red Imager
SKT	Skin Temperature
SLSTR	Sea and Land Surface Temperature Radiometer
SZA	Satellite (viewing) Zenith Angle
T2m	Temperature at 2 meters height
TCWV	Total Column Water Vapour
TIGR	Thermodynamic Initial Guess Retrieval
TIR	Thermal InfraRed

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TOA	Top-of-Atmosphere
Tskin	Skin temperature
UCSB	University of California Santa Barbara
UTC	Coordinated Universal Time
VIIRS	Visible Infrared Imaging Radiometer Suite
VR	Validation Report

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery take place in a timely manner and are complemented by the constitution of long-term time series.

This Algorithm Theoretical Basis Document (ATBD) describes the scientific background and the procedures used to calculate the global Land Surface Temperature (LST) obtained from a constellation of geostationary (GEO) satellite missions: Meteosat Second Generation (MSG) – 0° service and Indian Ocean Data Coverage (IODC) service; Geostationary Operational Environmental Satellite (GOES); and Himawari (and its predecessor Multi-Function Transport Satellite – MTSAT). These platforms are operated by different satellite agencies, but the data are made available in near real time via the EUMETSAT dissemination system, EUMETCast.

LST is the radiative skin temperature of land surface, as measured in the direction of the remote sensor, corrected for surface emissivity. Given the different characteristics of the imagers onboard each GEO platform, this document describes two sets of methodologies that can be applicable (and adjusted) to the available data. Both are based on semi-empirical functions that relate LST to top-of-atmosphere brightness temperatures in thermal (around 11 μm and 12 μm) and/or middle infrared (around 4 μm) window channels. The assumptions and physics underlying each methodology, as well as the uncertainties of LST estimates are discussed. The formulations are all trained using a dataset of radiative transfer simulations for a wide range of atmospheric and surface conditions. The performance of each algorithm is assessed by comparing its output against an independent set of simulations. The Generalized Split-Window (GSW) algorithm generally provides the lower uncertainty in LST retrievals when compared to the Dual-Algorithm (DA) method. However, its use is limited by the availability of brightness temperature measured in two adjacent channels within the thermal atmospheric window. Currently these are available in Near Real Time (NRT) for all the sensors used in the generation of the global CLMS LST product. However, users must take into account that for some of the sensors used in the past, this was not the case since: 1) the 12 μm channel was not available in the GOES sensors prior to GOES-16 and 2) in the case of MTSAT that channel existed but was not broadcasted in the EUMETCast system, and therefore was not available for the CGLOPS production chain. In those cases, the LST uncertainty was larger due to the usage of the DA algorithm, especially during daytime when only one channel is used to retrieve LST.

Surface emissivity is also a critical input of the processing chain. Within this product, it is estimated using the vegetation cover method, which weights bare ground and vegetation emissivities by the fraction of vegetation cover of each pixel. Vegetation and bare ground emissivities are prescribed for each land cover class, based on information derived from comprehensive spectral libraries, while the fraction of vegetation cover is based on the CLMS Fraction of vegetation Cover (FCOVER) 300m

product. Surface emissivity is also affected by the presence of snow, which is accounted for here using the CLMS Snow Cover Extent (SCE) 1km product, by replacing the pixel emissivities by those associated to the “snow / ice” land cover class, together with a correction for the viewing angle.

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1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The document includes the theoretical basis of the LST V2.2 retrieval methodology and details the physics background of the different algorithms, as well as expected accuracy, which are associated to the algorithms' uncertainties and to inputs errors, and limitations of the remote sensing data.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the user requirements
- Chapter 3 describes the methodologies that shall be used for LST estimation
- Chapter 4 summarizes some quality assessment results
- Chapter 5 presents the output LST product
- Chapter 6 identifies the risks and the mitigation measures
- Chapter 7 lists the references

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_FCOVER300m-V1.1	Algorithm Theoretical Basis Document of the FCOVER 300m Version 1.1 product
CGLOPS1_QAR_FCOVER300m-V1.1	Quality Assessment Report of the FCOVER 300m Version 1.1 product

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CGLOPS1_PUM_FCOVER1km-V2	Product User Manual of the FCOVER 1km Version 2 product
CGLOPS2_ATBD_SCE-NHEMI-1km_V1	Algorithm Theoretical Basis Document of the Northern Hemisphere Snow Cover Extent 1km V1.0 product
CGLOPS2_QAR_SCE-NHEMI-1km_V1	Quality Assessment Report of the Northern Hemisphere Snow Cover Extent 1km V1.0 product

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_LST-V2.2	Product User Manual summarizing all information about the LST Version 2.1 product
CGLOPS1_QAR_LST-LST10-V2.1	Quality assessment report of the LST product family Version 2.1

Available at: https://land.copernicus.eu/en/technical-library

1.3.4 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs requirements.

Available online at: https://library.wmo.int/

NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0	Scientific and Validation report for the Cloud Product Processors of the NWC/GEO, Issue 1, Rev. 0, 2016

Available online at:
http://www.nwcsaf.org/AemetWebContents/ScientificDocumentation/Documentation/GEO/v2016/NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0.pdf

JMA, 2017	Himawari-8/9 - Himawari Standard Data User's Guide, v1.3. Tokyo.

Available online at:
https://www.data.jma.go.jp/mscweb/en/himawari89/space_segment/hsd_sample/HS_D_users_guide_en_v13.pdf

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NOAA, 2019 GOES R Series Product Definition and Users' Guide. 416-R-PUG-L1B-0347 Vol 3 Rev.2.2.

Available online at: https://www.goes-r.gov/users/docs/PUG-L1b-vol3.pdf

S3-L2-SD-03-T03-ULNILU-ATBD_L2LST Sentinel-3 Optical Products and Algorithm Definition: SLSTR ATBD Land Surface Temperature, v2.3. European Space Agency.

Available online at: https://sentinel.esa.int/documents/247904/349589/SLSTR_Level-2_LST_ATBD.pdf)

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2 REQUIREMENTS

According to the applicable document [AD1], the specific technical details and requirements are described below (Table 1).

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

Table 1 – CGLOPS LST product specifications.

LST – LAND SURFACE TEMPERATURE	
The Land Surface Temperature (LST) is the radiative skin temperature of the land surface, as measured in the direction of the remote sensor [K].	
Geometric Properties:	
Spatial resolution	5 km
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global
Time span	Hourly and 10-day period
Timeliness	Hourly: within 4 hours; within 2 days (optimally 1 day) after the end of 10-day period
Uncertainty (2-sigma)	Threshold: 2 K, Goal: 1 K
Stability (per decade)	Threshold: 0.3 K, Goal: 0.1 K

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product;
- The per-pixel date of the individual measurement or the start-end dates of the period covered;
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result;

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NETCDF. Additional format, such as COG or ZAR, format can be proposed for production or requested by the contracting authority during the Framework.

2.2.4 Uncertainties and validation

Uncertainties indicated in the product specifications (Table 1) follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate, the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL datasets and/or Ground-Based Observations for Validation (GBOV, https://land.copernicus.eu/en/products/GBOV) of Copernicus Global Land component when biophysical parameters are available.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, and conversion to physical values formulas.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu;

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary datasets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 METHODOLOGY DESCRIPTION

3.1 OVERVIEW

Land Surface Temperature (LST) is estimated from Top-of-Atmosphere (TOA) brightness temperatures of atmospheric window channels within the infrared range. The algorithms developed for GEO satellites consider the available channels and are divided into three groups:

i) *split-window* methodologies, which make use of two adjacent window channels within the thermal infrared range (T_{TIR1} and T_{TIR2});

ii) *two-channel* algorithms, which derive LST from one window channel in the thermal infrared – around 11 μm – and another in the middle infra-red – around 3.9 μm (T_{TIR1} and T_{MIR}, respectively); two-channel algorithms are used when only one thermal infrared channel is available and for night-time conditions, when T_{MIR} is not contaminated by solar radiation reflected by the surface;

iii) *mono-channel* method that corrects the TOA brightness temperature of a single channel, T_{TIR1}, for atmospheric attenuation and surface emissivity; this algorithm is used for daytime conditions, when only one thermal infrared channel is available.

The methodologies mentioned above are all based on semi-empirical formulations, where LST is expressed as a regression function of TOA brightness temperatures. To minimize LST uncertainties, the algorithms are trained for different classes of satellite view angle, atmosphere water vapour content, and when needed, for different land cover types. A list of land cover types used in the product, derived from the IGBP database (Belward, 1996), is provided in Table 2.

In the current version of CLMS LST, global LST is derived using measurements from the Advanced Himawari Imager (AHI) onboard Himawari-8, the Advanced Baseline Imager (ABI) onboard GOES-16 and the Spinning Enhance Visible and Infrared Radiometer (SEVIRI) onboard the MSG (Meteosat Second Generation) platforms in charge of the 0º and IODC (Indian Ocean Data Coverage) missions. All these sensors allow the use of a generalized split window algorithm. Before GOES-16 became operational (December 2017), GOES used a two-channel approach for night-time and a mono-channel approach for day time, which is referred to hereafter as “Dual-Algorithm”. The same was true for Himawari disk area, which, until end of November 2015, used a similar approach since the second split-window channel from MTSAT was not available through EUMETCast. All these changes are summarized in Table 3.

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Table 2: Land cover type classification used in the product, based on IGBP classification (Belward, 1996).

Index / Code	Name	IGBP Land Cover Types
1 - Forest	Forest	1, 2, 3, 4
2 – MxFor	Mixed Forests	5
3 – ClShr	Closed Shrublands	6
4 – Savan	Savannas	7, 8, 9
5 – Grass	Grasslands	10
6 – Wetl	Permanent Wetlands	11
7 – CrpNV	Croplands and Natural Vegetation Mosaics	12, 14
8 – Urb	Urban and Built-Up	13
9 – SnowI	Snow and Ice	15
10 – Barren	Barren	16
11 – Water	Open water bodies	17
12 – InlWat	Inland water	18

Table 3: Chronology of the sensors used to produce global CLMS LST with indication of the retrieval algorithm.

<table>
<tr><th>Date</th><th>GOES-E</th><th>MSG 0º</th><th>MSG IODC</th><th>Himawari</th></tr>
<tr><td>10 Aug 2009</td><td rowspan="2">GOES Imager
(Dual-Algorithm)</td><td rowspan="4">SEVIRI
(Split-Windows)</td><td rowspan="2">Not available</td><td>MTSAT Imager
(Dual-Algorithm)</td></tr>
<tr><td>27 Nov 2015</td><td rowspan="3">AHI
(Split-Windows)</td></tr>
<tr><td>18 Dec 2017</td><td rowspan="2">ABI
(Split-Windows)</td><td rowspan="2">SEVIRI
(Split-Windows)</td></tr>
<tr><td>18 Jan 2020</td></tr>
</table>

3.2 THE RETRIEVAL ALGORITHM

3.2.1 Outline

The LST product is generated every hour (as illustrated in the workflow in Figure 1), through the following steps:

Step 1: Generate LST product for each GEO satellite disk (see the workflow in the Figure 2).

Step 1.1: Gather input data in the same resolution and projection as the satellite imagery;

Step 1.2: Decide which algorithm is to be used depending on the availability of the second thermal channel (TIR2);

Step 1.3: For all land pixels;

Step 1.3.1: In case of TIR2 is available for the given sensor, the LST is calculated with the Generalized Split Window (GSW) algorithm (section 3.2.5); surface emissivity is first estimated taking into account the Land Cover, Fraction of Vegetation Cover and Snow Cover of each pixel.

Step 1.3.2: In case of unavailability of TIR2 in the given sensor, check the solar elevation to decide whether the LST is to be calculated with or without the MIR channel i.e., by applying the two- or mono-channel algorithm, respectively (section 3.2.6);

Step 1.3.3: Calculate the uncertainties associated to the LST retrievals and set up a quality flag;

Step 2: Merge the LST (and the additional layers) for Himawari (or its predecessor MTSAT) and GOES disks with the LSA SAF-LST from both MSGs (0° and IODC missions) to obtain a global product.

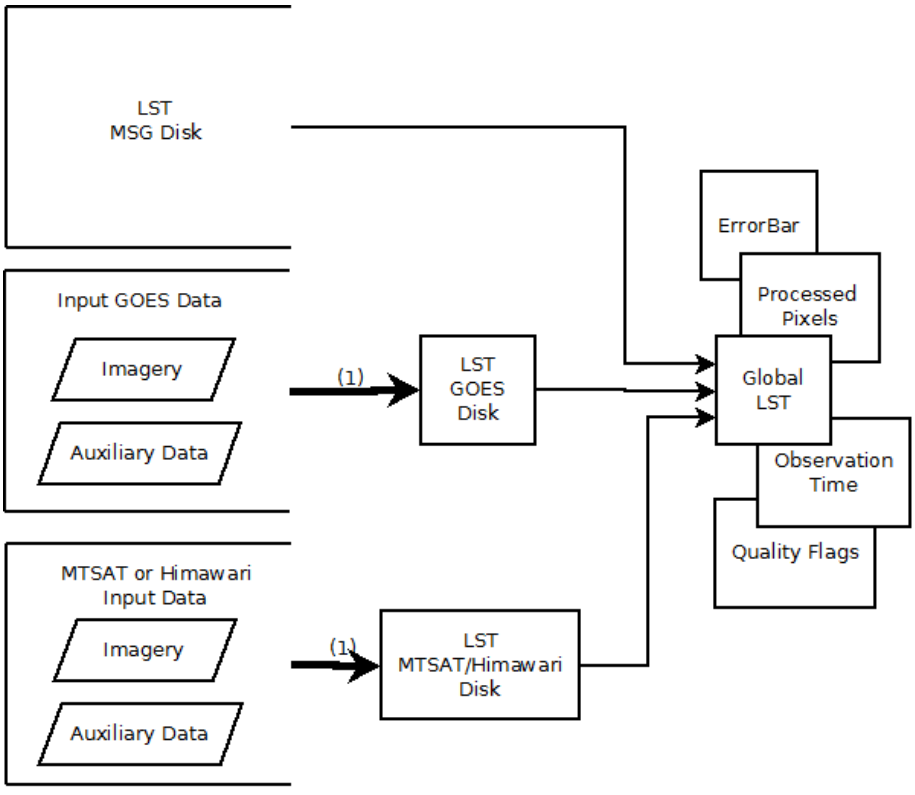


Figure 1: Flow diagram of the generation of Global LST and respective Error bars and Quality Flag. Data flow identified by (1) is detailed in Figure 2. MSG stands for both the 0º and the IODC missions in this context.

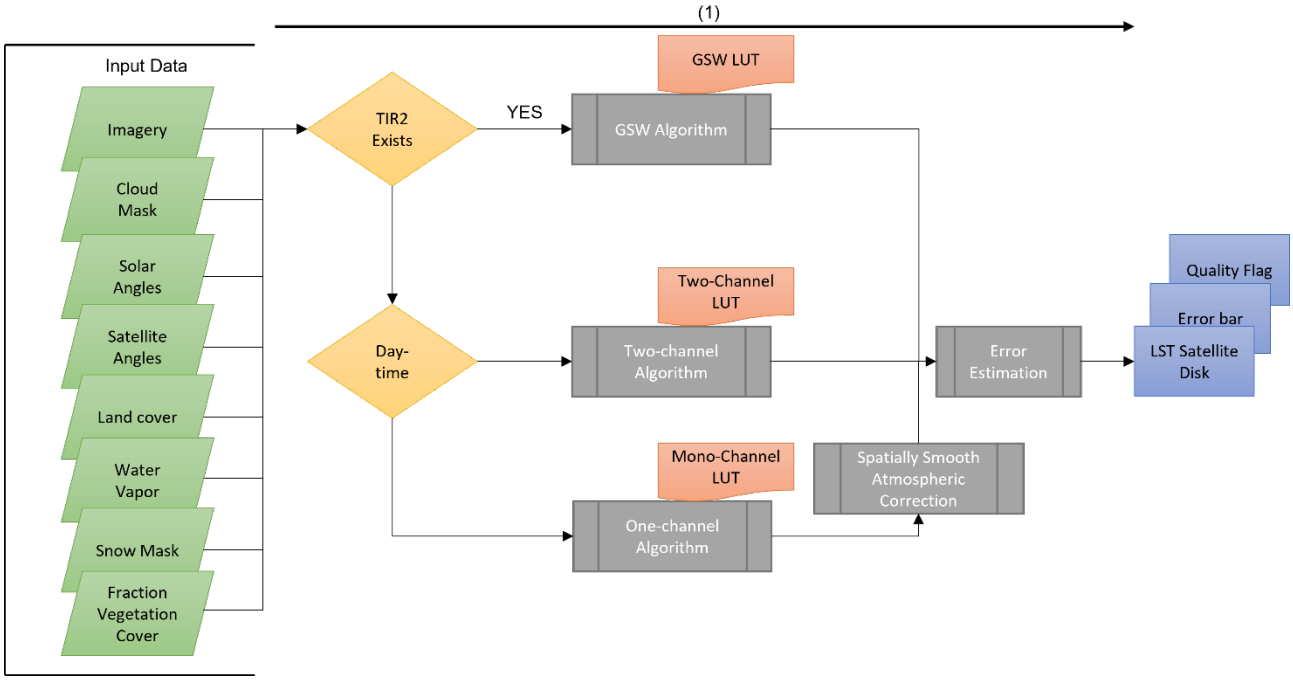


Figure 2: Flow diagram of the generation of LST for each GEO disk (Himawari or MTSAT & GOES).

3.2.2 Basic underlying assumptions

LST estimations from remotely sensed data are generally obtained from one or more channels within the thermal infrared atmospheric window from 8-to-13 μm (Dash et al., 2002, Li et al, 2013). Operational LST retrievals often make use of split-window algorithms (e.g., Prata, 1993; Wan and Dozier, 1996), where LST is obtained through a semi-empirical regression of top-of-atmosphere (TOA) brightness temperatures of two pseudo-contiguous channels, i.e., the split-window channels.

A relevant factor in the selection of the algorithm is its expected reliability for operational LST retrievals, both in terms of expected accuracy and timeliness. The latter favours the use of semi-empirical relationships between LST and TOA brightness temperatures, which are computationally efficient and free of the convergence problems of direct emissivity and temperature retrieval methods (e.g., Faysash and Smith, 1999; Hulley and Hook, 2011) associated to the non-linearity of the inverse problem in remote sensing (e.g., Rodgers, 2000). Recent studies have assessed the combination of middle infrared bands centred on 3.9 μm (T_{MIR}) with the split-window within the 10-to-13 μm band (T_{TIR1} and T_{TIR2}; Sun and Pinker, 2003, 2007; Pinker et al., 2007). Although there are several caveats regarding the use of such extra channels for LST operational retrievals, they can become a very useful source of information, particularly in the absence of two T_{TIR} bands. The major problems regarding the use of T_{MIR} include: (i) the uncertainty of surface emissivity within T_{MIR}, which is considerably higher than that of T_{TIR} channels, particularly over semi-arid regions (Trigo et al., 2008a); (ii) solar contamination of daytime T_{MIR} observations must be taken into account.

Ultimately, we want to generate a global LST field, which corresponds to the fusion of LST fields generated from four geostationary satellites: 1) GOES East, covering the Americas, 2) MSG4, covering most of Europe and Africa, 3) MSG1 covering the Indian Ocean, the Middle East and also viewing large parts of both Europe and Africa from the East, and 4) Himawari / MTSAT covering Eastern Asia and Oceania. The design of the algorithms applied to each individual sensor needs to take into account the channels described in Table 4 and Table 5.

Table 4: Satellite channels used for LST estimation using the GSW algorithm.

	MSG/SEVIRI	GOES/ABI	Himawari/AHI
TIR1	10.0 – 11.5 µm	10.8 – 11.6µm	10.8 – 11.6 µm
TIR2	11.2 – 12.8 µm	11.8 – 12.8µm	11.7 - 13.1 µm

Table 5: Satellite channels used for LST estimation using the DA algorithm

	GOES/Imager	MTSAT/JAMI
MIR	3.8 – 4.0 µm	3.5 – 4.0 µm
TIR1	10.2 – 11.2 µm	10.3 – 11.3 µm

Both ABI onboard GOES-16 (and following satellites of the same series) and AHI onboard Himawari-8 (and following satellites) provide three channels in the split-window region. The given combination of channels was chosen to achieve the best correspondence with SEVIRI/MSG and therefore facilitate product harmonization. The combination of the three (instead of the current two) ABI and AHI channels within the thermal infrared window to derive LST may be considered in a future version of the product (Yamamoto et al., 2018), e.g., upon the update to Meteosat Third Generation.

The GOES imager (from GOES-9 to GOES-13) provided measurements within a single TIR channel (Table 5). As a consequence, LST is estimated using observations from TIR1 and MIR during night-time and TIR1 alone during daytime, to avoid contamination from solar radiation reflected by the surface (Freitas et al., 2013). In the case of MTSAT, it actually provided observations for two split-window bands, but they were not disseminated via EUMETCast.

3.2.3 Related and previous applications

The LST algorithm for SEVIRI was developed by the EUMETSAT Satellite Application Facility (SAF) on Land Surface Analysis (LSA). The LSA-SAF generates, archives and disseminates

SEVIRI-based LST with a 15-minute frequency, at the original satellite spatial resolution. The Copernicus Global Land LST product has benefited from the experience gained during the development of the LSA-SAF LST product for the design and implementation of GOES, MTSAT and Himawari LST algorithms.

To overcome some of its known limitations, the product has suffered some evolutions since the beginning of the operational service. Table 6 shows the evolution of the product versions, together the main associated changes.

Table 6 : List of product versions with main associated changes

Version	Change	Date
V1.0	Initial version.	
V1.1	Improved atmospheric correction after detection of artifacts	
V1.2	Algorithm updated to include Himawari retrievals and spatial resolution changed to 5/112°. All data from beginning of timeseries was reprocessed with this version.	19.06.2010 to 18.01.2021
V1.3	Fix coordinate reference system (crs) inverse_flattening attribute and global attributes “sensor”,platform” All data from beginning of time series was reprocessed with these fixes. No change in the actual products content.	19.06.2010 to 18.01.2021
V2.0	Emissivity is now seasonally varying and is estimated using the Vegetation Cover Method with CLMS FCOVER 1 km as input. A common calibration database was introduced for both AHI/Himawari and ABI/GOES16, so GSW model coefficients were updated for both sensors with respect to previous product versions. MSG IODC mission was introduced in the product, covering the Middle East. The timeslot for MSG 0º is now T=0 (whereas in previous versions the T-15 min was used).	18.01.2021 to 31.12.2022
V2.1	Snow cover is considered in the estimation of surface emissivity for Himawari and GOES-16 missions. Sentinel-3 based CLMS FCOVER 300m is now used as input.	01.01.2023 to 02.06.2026
V2.3	Fix coordinate reference system (crs) inverse_flattening attribute and global attributes “sensor”,platform”	18.01.2021 to near real time

	All data from 18.01.2021 was reprocessed with these fixes. No change in the actual products content.	

3.2.4 Alternative methodologies currently in use

A thorough overview of algorithms developed for the estimation of LST from remote sensing observations within the infra-red domain may be found in Li et al., 2013. Satellite-based thermal infrared (TIR) data is directly linked to the LST through the radiative transfer equation. However, the estimation of LST from satellite TIR measurements requires correction for surface emissivity and for atmospheric effects. The algorithms used for operational production of LST are generally designed to use information from all channels that may help characterizing surface emission and atmospheric correction, while minimizing product uncertainty and generation time. When split-window bands are available, generalized split-window methods fulfil such prerequisites and therefore are the most widely used for operational LST products, e.g.:

- MODIS daily products MOD11A1 (MODIS on Terra) and MYD11 (MYD11A1) are based on the algorithm developed by Wan and Dozier (1996);
- a similar algorithm is used by the LSA-SAF for SEVIRI-based LST;
- the AATSR/Envisat LST distributed (off-line) by ESA is based on a split-window formulation developed by Prata (1993);
- the SLSTR Level-2 LST from the ESA Sentinel-3 mission, which uses a simpler split-window formulation described in [S3-L2-SD-03-T03-ULNILU-ATBD_L2LST];
- the NOAA LST operational product retrieved from VIIRS/NPP, also using a generalized split-window algorithm (Yu et al, 2005).

The split-window algorithms assume that land surface temperature is linearly related to the TOA brightness temperature within the TIR range, apart from an atmospheric and surface emissivity correction. The atmospheric correction is a function of the differential absorption of the two adjacent channels TIR1 and TIR2. There are several formulations of generalized split-window algorithms. Yu et al. (2008) compared some of the most used documented formulations (Table 7).

The performance of LST algorithms often depends on the retrieval conditions, as further detailed in section 3.2.5.2, and on the sensitivity to input uncertainties. As an example, Figure 3 presents the LST error budget estimated for the set of 9 split-window algorithms shown in Table 7 assuming realistic uncertainties in the respective input parameters and viewing angle up to 22º. The generalized split-window corresponds to algorithm #1. Figure 3 clearly shows that most formulations tend to perform worse under moist atmospheres (and with increasing viewing angle, although not shown). The cluster of methods with overall lower errors has in common an explicit correction of surface emissivity and coefficients tuned to classes atmospheric water vapor content and view angle. The differences in performance among these, which include the Wan and Dozier (1996), are very small.

Table 7: Split-window formulations tested in Yu et al. (2008) and their respective references. T_S denotes *LST*; C, A_1, A_2, A_3, B_1, B_2, B_3 are the regression coefficients, T_{11} and T_{12} are the brightness temperatures in the IR 108 and IR 120 channels, ϵ_{11} and ϵ_{12} their emissivities and ϵ the average of both emissivities.

N°	Formula	Reference(s)
1	$T_s = C + \left(A_1 + A_2\frac{1-\epsilon}{\epsilon} + A_3\frac{\Delta\epsilon}{\epsilon^2}\right)\frac{T_{11}+T_{12}}{2} + \left(B_1 + B_2\frac{1-\epsilon}{\epsilon} + B_3\frac{\Delta\epsilon}{\epsilon^2}\right)\frac{T_{11}-T_{12}}{2}$	(Freitas *et al.*, 2010; Wan and Dozier, 1996)
2	$T_S = C + A_1\frac{T_{11}}{\epsilon} + A_2\frac{T_{12}}{\epsilon} + A_3\frac{1-\epsilon}{\epsilon}$	(Caselles *et al.*, 1997; Prata and Platt, 1991)
3	$T_S = C + A_1T_{11} + A_2(T_{11}-T_{12}) + A_3(1-\epsilon) + A_4\Delta\epsilon$	(Ulivieri *et al.*, 1994)
4	$T_S = C + A_1T_{11} + A_2(T_{11}-T_{12}) + A_3\frac{1-\epsilon}{\epsilon} + A_4\frac{\Delta\epsilon}{\epsilon^2}$	(Vidal, 1991)
5	$T_S = C + A_1T_{11} + A_2(T_{11}-T_{12}) + A_3(T_{11}-T_{12})(1-\epsilon_{11}) + A_4T_{12}\Delta\epsilon$	(Price, 1984)
6	$T_S = C + A_1T_{11} + A_2(T_{11}-T_{12}) + A_3\epsilon$	(Ulivieri and Cannizzaro, 1985)
7	$T_S = C + A_1T_{11} + A_2(T_{11}-T_{12}) + A_3\epsilon + A_4\frac{\Delta\epsilon}{\epsilon}$	(Sobrino *et al.*, 1994)
8	$T_S = C + A_1T_{11} + A_2(T_{11}-T_{12}) + A_3(1-\epsilon_{11}) + A_4\Delta\epsilon$	(Coll *et al.*, 1997)
9	$T_S = C + A_1T_{11} + A_2(T_{11}-T_{12}) + A_3(T_{11}-T_{12})(T_{11}-T_{12}) + A_4(1-\epsilon_{11}) + A_5\Delta\epsilon$	(Sobrino *et al.*, 1993)

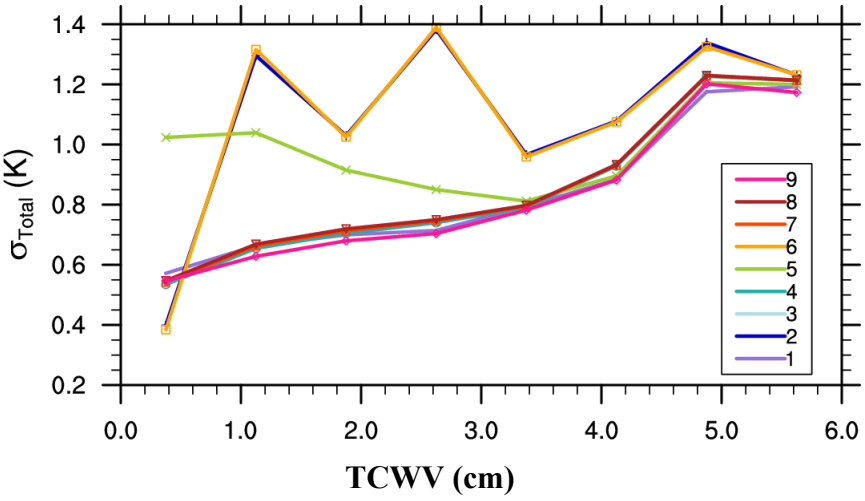


Figure 3: Error budget estimated for 9 different formulations of split window algorithms (discussed in Yu et al., 2008), taking into account the uncertainty of the split-window regression and of input data (sensor noise, emissivity and total column water vapour).

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3.2.5 Generalized Split-Window Algorithm

3.2.5.1 Input data

The Generalized Split-Window algorithm (GSW) provides estimation of LST using measurements from two adjacent channels within the infrared atmospheric window, T_{TIR1} and T_{TIR2} (Table 4).

All inputs are re-gridded (both in time and space, when needed) for each pixel in the original satellite geostationary projection, so that all algorithm inputs are in a common grid. In the case of GOES-R, the radiance quality flag is used to ensure that only pixels with "good" quality are used because pixels with less-than-ideal quality can cause outliers and unrealistic LSTs.

The GSW requires the following inputs:

- Land Cover data base - provided by the International Geosphere-Biosphere Program (IGBP) database (Belward, 1996). All valid pixels of the input dataset falling within each pixel of the satellite projection are selected, and the land cover type attributed to the pixel corresponds to the most frequent class.
- Fraction of Vegetation Cover for
    - SEVIRI daily pixels – estimated by the LSA-SAF from surface reflectance data (García-Haro *et al.*, 2005; Roujean *et al.*, 1992; Trigo *et al.*, 2011)
    - Other sensors – Copernicus Global Land Service FCOVER Collection 300m product [CGLOPS1_ATBD_FCOVER300m-V1.1];
- Lookup tables for the bare soil and vegetation emissivities, for each land cover class, and for each TIR channel of each sensor
- Snow fraction, derived from CLMS Snow Cover Extent 1 km product for Northern Hemisphere [CGLOPS2_ATBD_SCE-NHEMI-1km_V1] (Metsämäki et al., 2012, 2015)
- Cloud mask for:
    - SEVIRI – obtained from the Nowcasting and Very Short-Range Forecasting (NWC) SAF software
    - Other sensors – developed by the project team, based on the NWC SAF software (see Annex 3: Cloud mask)
- Total Column Water Vapor (TCWV), which is obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF) operational forecasts, retrieved with an hourly frequency and at 9km resolution. Forecast fields are bilinearly interpolated to the satellite grid for the neighbouring time steps, and then linearly interpolated in time to the acquisition time of each pixel.

MSG-based LSTs (derived over the 0° nominal disk and IODC) are processed by the LSA-SAF and then directly ingested in the CLMS global product, while GOES and Himawari LST fields have specific CGLOPS processing chains. The LSA SAF LST uses the LSA-SAF FVC product (García-Haro *et al.*, 2005; Roujean *et al.*, 1992; Trigo *et al.*, 2011), derived in the same processing chain and from the same sensor (SEVIRI) on a daily basis. Since the LSA-SAF FVC is limited to the MSG disks, LST retrievals from GOES and Himawari missions make use the CLMS FCOVER 300m V1.1 [CGLOPS1_ATBD_FCOVER300m-V1.1]), which corresponds to the same variable (i.e., fraction of green vegetation per pixel). As shown in Annex 4: Migration to CLMS FCOVER 300m, it is clear that the LSA SAF and CLMS FCOVER provide very similar estimates for the superposition zone between MSG and GOES, with differences mostly within the [-0.1, 0.1] interval, i.e., they are essentially within the uncertainty most frequently estimated for the LSA-SAF fraction of vegetation cover product, yielding emissivity uncertainties generally below 0.001 [see also CGLOPS1_QAR_LST-LST10-V2.1]. Therefore, despite the use of different vegetation sources, the impact on the final emissivity estimate is small, while any of those vegetation products serve their purpose of imprinting a seasonal cycle on emissivity and discriminating between sparsely and densely vegetation surfaces. Moreover, FVC changes relatively smoothly, unless there is some catastrophic event such as a wildfire or a flood. So, in general we do not expect any major impacts due to the use of a decadal (CLMS) instead of a daily (LSA-SAF) vegetation product.

3.2.5.2 LST retrieval

The GSW algorithm, used for example at LSA-SAF in their LST products (Trigo et al., 2008b), uses a formulation similar to the one proposed by Wan and Dozier (1996) for AVHRR and MODIS. In this algorithm, LST is as a function of TOA brightness temperatures of each sensor split-window channels (T_{TIR1} and T_{TIR2}, respectively) – see Table 4):

$$LST = \left(A_1 + A_2\frac{1-\epsilon}{\epsilon} + A_3\frac{\Delta\epsilon}{\epsilon^2}\right)\frac{T_{TIR1}+T_{TIR2}}{2} + \left(B_1 + B_2\frac{1-\epsilon}{\epsilon} + B_3\frac{\Delta\epsilon}{\epsilon^2}\right)\frac{T_{TIR1}-T_{TIR2}}{2} + C + \Delta LST \quad (1)$$

where ε is the average of the two channels surface emissivities, Δε their difference (ε_{TIR1} - ε_{TIR2}), while A_j, B_j, (j = 1,2,3) and C are the GSW coefficients obtained by fitting equation (1) to a calibration dataset for different classes of water vapour, TCWV, and satellite or viewing zenith angle (SZA), and ΔLST is the model error. The calibration and verification of the model imply the realization of a set of radiative transfer simulations of TOA brightness temperature values for a wide range of surface, atmospheric and geometric conditions, which are then separated into a calibration dataset and a verification dataset (fully described below). The model error, ΔLST, may be assessed by comparing retrieved LST values to the "true" values, which were used to force the radiative transfer simulations in the verification database (which is similar, but independent of the database used to calibrate the algorithm). Note that those comparisons only assess the algorithm performance, neglecting all the

input uncertainties and their propagation into the final LST estimate. By performing those comparisons, it is possible to assess if, once the model is calibrated, it can reproduce the LSTs serving as input to radiative transfer simulations in the verification database, using the brightness temperatures resulting from those simulations as input. Therefore, it is not a validation, since that procedure would imply the comparison of real retrievals against independent measurements of the retrieved variable, such as those provided by in situ stations. The GSW algorithm is applied to clear sky pixels only (Freitas et al., 2010). The pixel emissivity for channel *i* is estimated as a function of the fraction of vegetation cover (*FVC*) according to the equation (Becker and Choudhury, 1988; Caselles *et al.*, 1997):

$$\boldsymbol{\epsilon_i = FVC \times \epsilon_{veg} + (1 - FVC) \times \epsilon_{BS}}, \quad (2)$$

where ε_{BS} and ε_{Veg} represent the emissivities of bare soil and vegetation, respectively. The values of the emissivities (ε_{BS} and ε_{Veg}) and their uncertainties are obtained for each sensor and for each channel using a spectral emissivity library of surface materials and convolving each individual spectrum with the channel response function and then averaging for all materials within a given land cover type (Peres and DaCamara, 2005; Trigo *et al.*, 2008b). The spectral emissivity library is composed of data obtained from the Johns Hopkins University (JHU) (Salisbury and D'Aria, 1992) and Jet Propulsion Laboratory spectral libraries included in the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) spectral library, as well as from the Moderate Resolution Imaging Spectrometer (MODIS) University of California—Santa Barbara (UCSB) spectral library. Most used information is from JHU, which is based on figures from laboratory measurements of the spectral reflectance of different types of materials. The aggregation of spectra in the different land cover classes is detailed in Peres and DaCamara (2005). The uncertainty here is the absolute uncertainty, i.e. the largest absolute deviation from the mean emissivity within each class. Each spectrum in the spectral database is convolved with the channel response function for each sensor to obtain channel emissivities for purely bare soil and vegetation types. Spectral response functions were obtained from https://www.nwpsaf.eu/site/software/rttov/download/coefficients/spectral-response-functions/.
These values are shown in Table 12 to Table 14 in Annex 1: Emissivities and their uncertainties.

FVC is estimated dynamically, but provided by different sources. The LSA-SAF missions (MSG 0° and IODC) use their own daily FVC product, based on SEVIRI observations (García-Haro *et al.*, 2005; Roujean *et al.*, 1992; Trigo *et al.*, 2011). In the cases of GOES and Himawari missions, the decadal CLMS FCOVER 300m V1.1 product, based on Ocean and Land Color Instrument (OLCI) data onboard Sentinel-3 platform, is used [CGLOPS1_ATBD_FCOVER300m-V1.1]. It is integrated in the CGLOPS LST processing chain as follows: first, the FCOVER 300 m product is resampled to the satellite projection corresponding to each mission, and then all the valid FCOVER grid points contained in each pixel of the GEO grid are averaged to obtain the final pixel estimate. The same applies for the FCOVER product uncertainty. These are then used as inputs for the calculation of the channel emissivities (cf. eq. 2) and their uncertainties (eq. 18 below). Because the FCOVER NRT estimates present many missing data, it was decided to use the consolidated mode after one dekad

(RT1) which is the best compromise between product completeness and up-to-date information [see Figure 57 of CGLOPS1_QAR_FCOVER300m-V1.1]. However, RT1 fields are not spatially complete, especially over regions with frequent cloud cover (e.g., the tropical rainforest). In those cases, a pre-computed "climatology" for each dekad was derived from the PROBA-V-based FCOVER 1km Version 2 (data from 2014-2020) [CGLOPS1_PUM_FCOVER1km-V2], which is used to fill in the gaps of the NRT product, thus ensuring that a realistic, seasonally varying value is used in any case. This "climatology" is also used as a backup in case there is a failure in the reception of the NRT FCOVER data. Indeed, the OLCI-based FCOVER product is highly consistent with the PROBA-V based FCOVER 1km Version 2 products as demonstrated in the Quality Assessment Report [CGLOPS1_QAR_FCOVER300m-V1.1].

From product V2.1 onwards, the GOES and Himawari retrievals are further corrected to account for the presence of non-permanent snow. This was already done in the case of the MSG 0º and IODC disks, as LSA-SAF already used an internal snow mask derived from albedo measurements obtained by SEVIRI. In the cases of the GOES and Himawari retrievals, the CLMS Snow Cover Extent 1 km V1.0 product [CGLOPS2_ATBD_SCE-NHEMI-1km_V1] is used to estimate a snow fraction for each pixel. The 1 km data provides an estimate of the fraction occupied by snow over each grid point. These are used to estimate the average snow cover over each pixel of the GEO disk, similarly to what is done for FCOVER. A pixel is then considered as containing snow if its averaged snow fraction is not zero. In this case, the algorithm considers the pixel is composed by two fractions: the snow-covered fraction and the non-snow-covered fraction, to which channel emissivities are computed (ϵ_{snow} and ϵ_{nosnow}, respectively). ϵ_{BS} and ϵ_{veg} values for land cover type "Snow / Ice" are used to estimate the snow-covered fraction emissivities. The final channel emissivity is given by:

$$\boldsymbol{\epsilon = SCE \times \epsilon_{snow} \times f(SZA) + (1 - SCE)\epsilon_{nosnow}}, \quad (3)$$

where SCE is the snow cover extent. The snow-covered fraction contains a term to correct for angular variations of surface emissivity. To obtain the angular dependence, a rough-surface specular bidirectional reflectance distribution function (BRDF) model is applied to each snow/ice sample in the spectral library to obtain angle varying estimates of channel emissivities, following Peres and DaCamara (2005), with integrated kernel values taken from Snyder and Wan (1998). Emissivity spectra were estimated for SZA angles between 0 and 60° and then convoluted with the sensors' response functions to obtain channel values. The angular dependent emissivities for snow is shown in Figure 4. These values are prescribed to the processing chain through a LUT (see Table 15) and linearly interpolated to each pixel's SZA.

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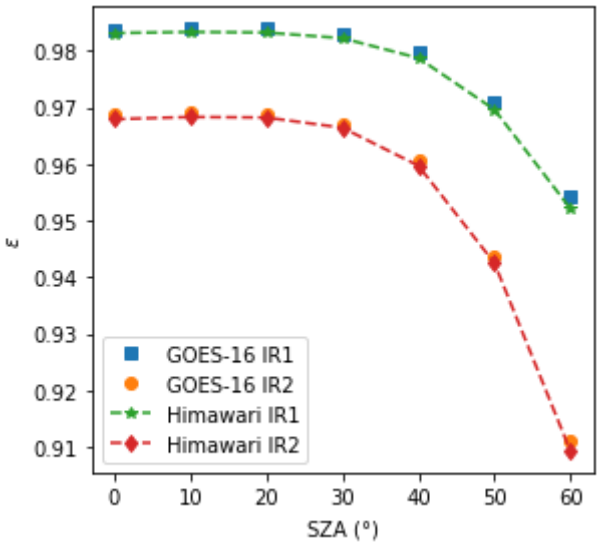


Figure 4: Angular variation of surface emissivity for channels IR1 and IR2, for Himawari and GOES-16.

Since snow cover extent can vary rapidly (both in time and space) through snow fall or snow melting, we did not consider the use of a "climatological" backup beyond that already implicit in the land cover classification. Furthermore, CLMS SCE data is only available from 2018 onwards, so the record is still too short to provide a robust and realistic seasonal field. Therefore, in the absence of NRT SCE information, the original land cover classification is used (which includes areas of permanent snow-ice). Figure 5 exemplifies the impact of this change for one timeslot (2021-01-15 00UTC). Because the Snow Cover Extent product only covers the Northern Hemisphere and is only used over the GOES and Himawari domains, this change will only impact East Asia and North America. The bottom panel in Figure 5 shows the distribution of the LST differences over one full day and for the full CLMS LST domain (zero values are excluded from the distribution, i.e., only pixels where snow was identified by the SCE product are considered). In general, the LSTs become up to 2 K colder in most situations, with a sharp peak between -1.2 and 1.6 K. A thorough discussion on the impact of including snow cover information on the LST retrieval is included in the Quality Assessment Report [CGLOPS1_QAR_LST-LST10-V2.1].

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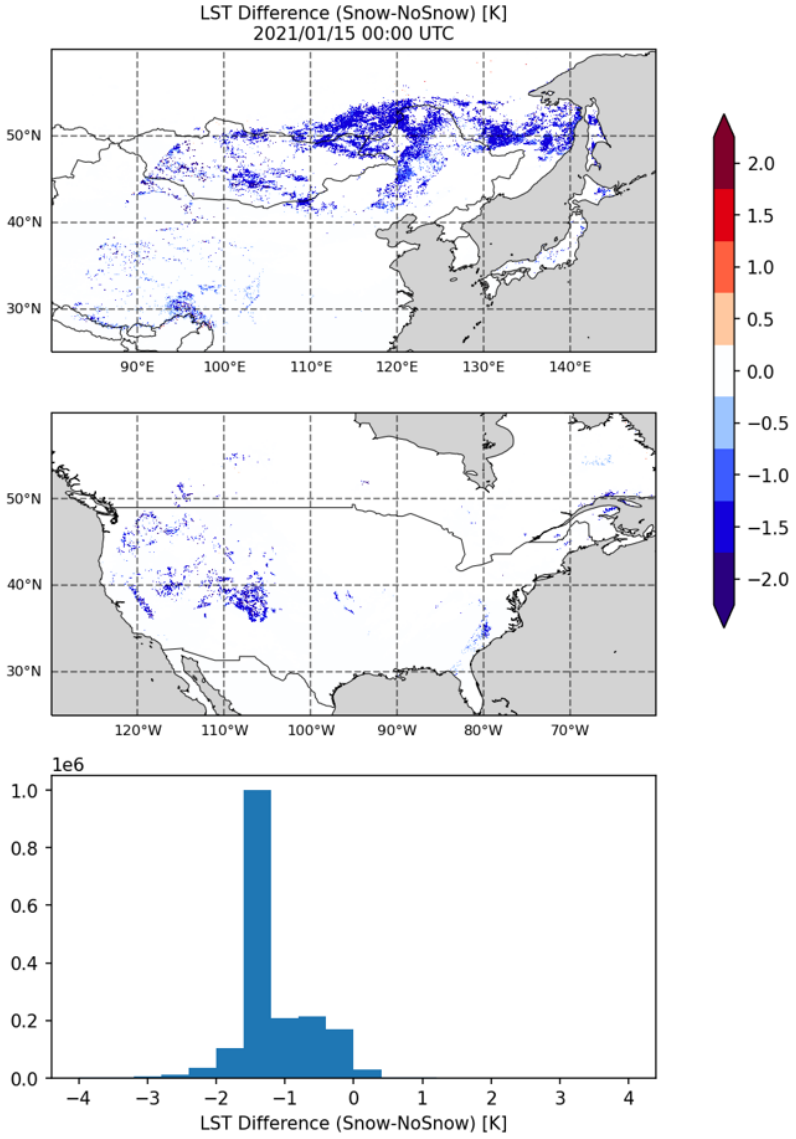


Figure 5: Impact of snow for East Asia (top) and North America (center) for the 20210105 00UTC timeslot. In the bottom, the LST difference distribution for all the timeslots in that day is shown.

The calibration and verification databases of the GSW developed for each sensor rely on radiative transfer simulations of TOA brightness temperatures for the split-window channels available in the sensors considered here. The simulations are performed for the database of global profiles of temperature, moisture, and ozone compiled by Borbas et al. (2005) for clear sky conditions and referred hereafter as SeeBor. The database contains over 15,700 profiles taken from other datasets, such as NOAA88 (Seemann et al., 2003), Thermodynamic Initial Guess Retrieval (TIGR)-like (Chevallier, 2001), and TIGR (Chedin et al., 1985), that are representative of a wide range of atmospheric (clear sky) conditions over the whole globe. In addition, surface parameters such as skin temperatures (Tskin) and a landcover classification within the International Geosphere-Biosphere

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Programme ecosystem categories (IGBP) (Belward, 1996) are assigned to each profile. Skin temperature over land surfaces corresponds to LST in SeeBor and is estimated as a function of 2m temperature (T2m), biosphere information, and solar zenith and azimuth angles (Borbas et al. 2005). In the design of the calibration database, random viewing angles within the range of acceptable usage are assigned to each profile – viewing zenith angle values above 70º are not used in the CGLOPS LST merged product (see section 3.3), as the signature of surface emitted radiation (and therefore LST) weakens for very high viewing angles.

The calibration methodology used here follows the study by Martins et al. (2016). The SeeBor database described above was split into two subsets – one used for the calibration of the GSW algorithms, and an independent one used for verification of the fitted versions. The calibration subset strongly influences the robustness of the model coefficients; therefore, the choice of the profiles requires some caution. The SEVIRI algorithm was calibrated within the framework of LSA-SAF using 77 geographically well distributed atmospheres, covering a broad variety of water vapour content (from very dry to moist conditions), leaving more than 15,600 profiles for GSW verification (Freitas et al., 2010); this corresponds to the algorithm currently used operationally to generate LSA-SAF LST products. The calibration of ABI and AHI LST algorithms made use of a new and more representative selection of profiles, taken for areas outside the MSG disk, which lead to a lower regression error (Martins et al., 2016). The composition of such calibration database follows the steps below:

i) Define classes of T_{Skin} (from 200 K to 330 K in steps of 5 K) and TCWV (from 0 to 6 cm in classes of 0.75 cm—values greater than this should be treated with the coefficient corresponding to the last TCWV class)

ii) Iterate in the SeeBor clear-sky profile database to fill each class in the TCWV/TSkin phase space (as in Figure 6c) with one case each. When a new profile is selected, it is ensured that its great-circle distance to the already selected profiles is greater than an initial distance of 15 degrees, which guarantees a wide geographical coverage. After a sufficiently large number of tries (in this case 30,000), the distance criterion is relaxed in steps of minus 1 degree, until the whole TCWV/TSkin phase space is filled. The final set has 116 profiles.

iii) For each of the previously selected profiles, assign a new T_{Skin} based on the ranges of $T_{Skin} - T_{air}$ observed range of this difference. The choice of the range of perturbations to apply is key to the performance of the chosen model and may depend on the region of interest. In the case of this work, a range of ±15K around T_{air} in steps of 5K showed an overall good performance. As will be seen, large biases arise when non-physical cases are included or if the somewhat more extreme cases are not taken into account.

iv) Each of these conditions may be sensed from angles ranging from 0 (nadir view) to 70° in steps of 2.5°. It is important to discretize the viewing geometry in this way because this is

an intrinsically non-linear problem. The upper limit of the viewing zenith angle might be adapted for the sensor under analysis. Previous calibration exercises show that above this viewing angle limit the retrieval errors are generally too high (i.e. above 4 K, the threshold uncertainty for CLMS LST), especially for moister atmospheres (Freitas et al., 2010);

v) For the emissivity, a range of possible values are attributed to each of the cases above: values of ϵ_{TIR1} from 0.93 to 1.0 in steps of 0.01 and then prescribe departures from this value for ϵ_{TIR2}: -0.015 to 0.035 in steps of 0.01 (excluding cases where $\epsilon_{TIR2} > 1$), as suggested by the distribution of this difference within the SeeBor dataset.

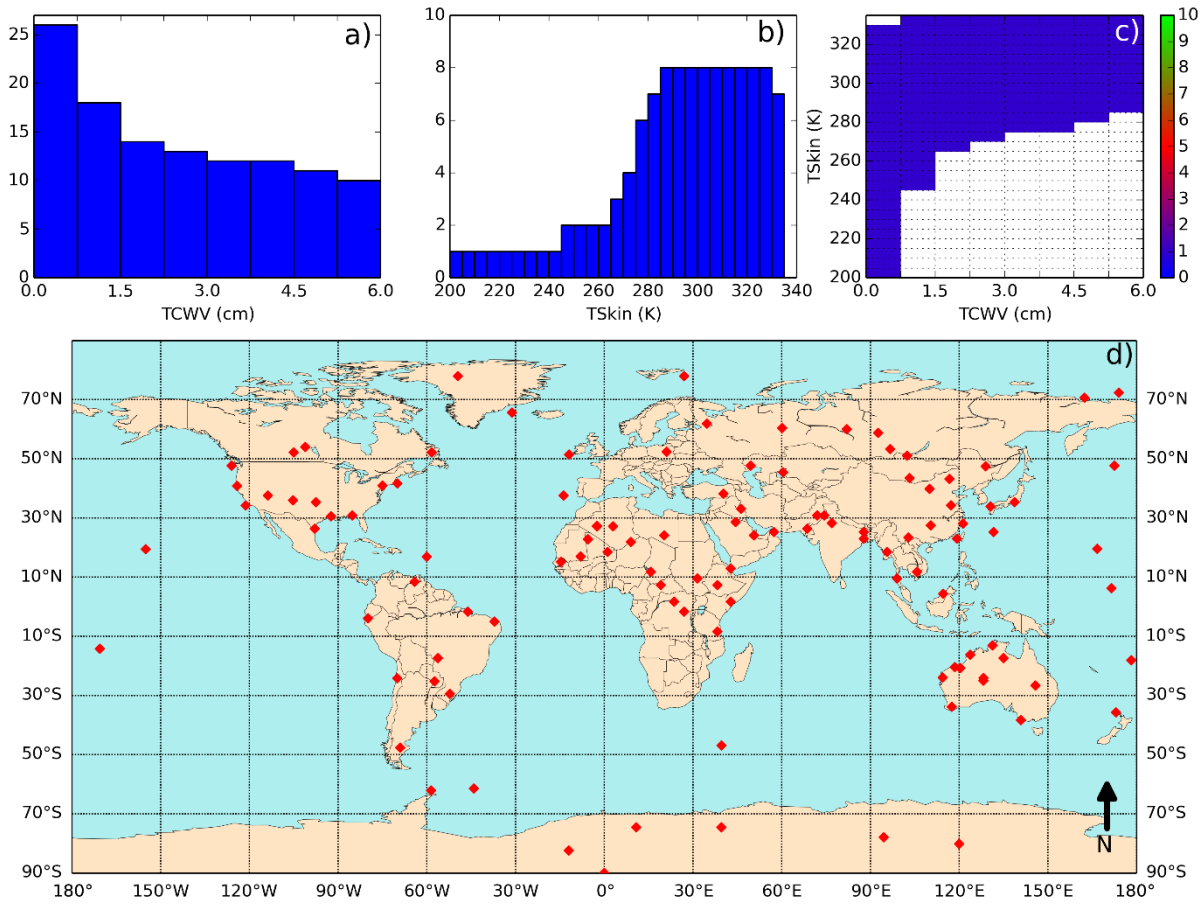


Figure 6: Main properties of the calibration database atmospheric profiles used in the determination of the model coefficients for GOES and Himawari. (a) TCWV distribution; (b) T_{Skin} distribution; (c) bivariate $TCWV/T_{Skin}$ distribution and (d) geographical distribution.

The radiative transfer simulations are performed using the MODerate spectral resolution atmospheric TRANsmittance algorithm (MODTRAN6; Berk et al., 2014). The radiance (L_ν) that would be measured by each TIR channel is then estimated with MODTRAN6, for each of the surface and

atmospheric conditions, and viewing geometry described in the previous paragraph. The simulations are performed using a spectral resolution of 1 cm^{-1}. The integration of L_ν weighted by the *i*-th channel response function $\phi_{i,\nu}$, provides channel *i* effective radiance:

$$L_i = \frac{\int_{\nu_{i_1}}^{\nu_{i_2}} \phi_i(\nu) L_\nu \, d\nu}{\int_{\nu_{i_1}}^{\nu_{i_2}} \phi_i(\nu) \, d\nu} \tag{4}$$

where $\nu_{i,1}$ and $\nu_{i,2}$ are the lower and upper wave number boundaries of the channel, respectively; the integrals in (3) are estimated taking into account tabulated values of response function of each channel. Those values are then subject to the inverse Planck function, B^{-1}, to obtain an estimate of the channel effective brightness temperatures, T^*_{IR1} and T^*_{IR2} (e.g., Freitas *et al.*, 2013):

$$T_i^* = B^{-1}(\nu_{i,c}, L_i) = \frac{c_2 \nu_{i,c}}{\log\left(\frac{c_1 \nu_{i,c}^3}{L_i} + 1\right)} \tag{5}$$

where $\nu_{i,c}$ is the central frequency of each channel (in cm^{-1}), $c_1 = 1.19104 \times 10^{-5}$mW.cm^2 and $c_2 = 1.43877$ K.cm are constants. However, these are the effective brightness temperatures estimated using $\nu_{i,c}$ without taking into account that the radiance L_i corresponds to the whole band, following the instrument response function. A further correction is then required to convert T_i^* into spectral brightness temperature, T_{b_i}. The latter is channel/ sensor dependent and each satellite agency uses slightly different methodologies to perform this conversion. In the case of GOES and according to the Product User Guide [NOAA, 2019], it takes the form:

$$\boldsymbol{T_{b_i}} = \frac{\boldsymbol{T_i^* - bc_{1_i}}}{\boldsymbol{bc_{2_i}}} \tag{6}$$

where fk_{1_i}, fk_{2_i}, bc_{1_i} and bc_{2_i} are the coefficients which are calculated taking into account each channel response function and are provided within the metadata of the products disseminated via EUMETCast. In the case of Himawari, the Standard Data User's Guide [JMA, 2017] states that the correction is performed using a 2nd degree polynomial expression of the form:

$$\boldsymbol{T_{b_i} = b_{1_i} + b_{2_i} T_i^* + b_{3_i} T_i^{*2}} \tag{7}$$

Where, again, b_{1_i}, b_{2_i} e b_{3_i} are the band coefficients provided in the input file metadata.

For SEVIRI on MSG 0° and IODC missions, the performed correction is linear as in the case of Himawari (https://www-cdn.eumetsat.int/files/2020-04/pdf_effect_rad_to_brightness.pdf):

$$\boldsymbol{T_{b_i} = (T_i^* - \beta)/\alpha} \tag{8}$$

where α and β are the band correction coefficients.

The GSW parameters A_i, B_i and C obtained by fitting equation (1) to the calibration dataset and the variance of LST explained by the regression are schematically shown in Figure 7 (example for

SEVIRI). The calibration is performed offline, and the coefficients are kept in look-up-tables used by each sensor. The ranges for the look-up tables are the same as the ranges for the simulations. The GSW algorithms are verified against the independent subset of simulated TOA brightness temperatures (which excludes the calibration data). The fact that sometimes coefficients do not vary monotonically reflect the non-linear nature of the retrieval problem and justify the need to calibrate the algorithm using discrete classes of SZA and TCWV. Non-linearities mainly rise from the fact that there is some compensation between surface emission and the reflected component of the atmospheric emission in the total radiance observed by the satellite. For drier atmospheres, transmissivity in the atmosphere is higher and longwave emission from the atmosphere is lower. In these cases, there is a high sensitivity to emissivity (see section 3.2.5.3.2 below). For moister atmospheres, the emission from the atmosphere increases and transmissivity decreases. Moreover, the surface upward longwave flux has two components: 1) the surface emission component (the signal we are interested in), which is proportional to ϵ; and 2) the reflected atmospheric radiance component, which is proportional to $1 - \epsilon$, meaning that its impact on the retrieval is minimized for higher emissivities. All these effects modulating the radiance measured by the satellite have non-linear interactions; otherwise, it should be possible to model these dependencies with a simple linear function of SZA, TCWV and ϵ.

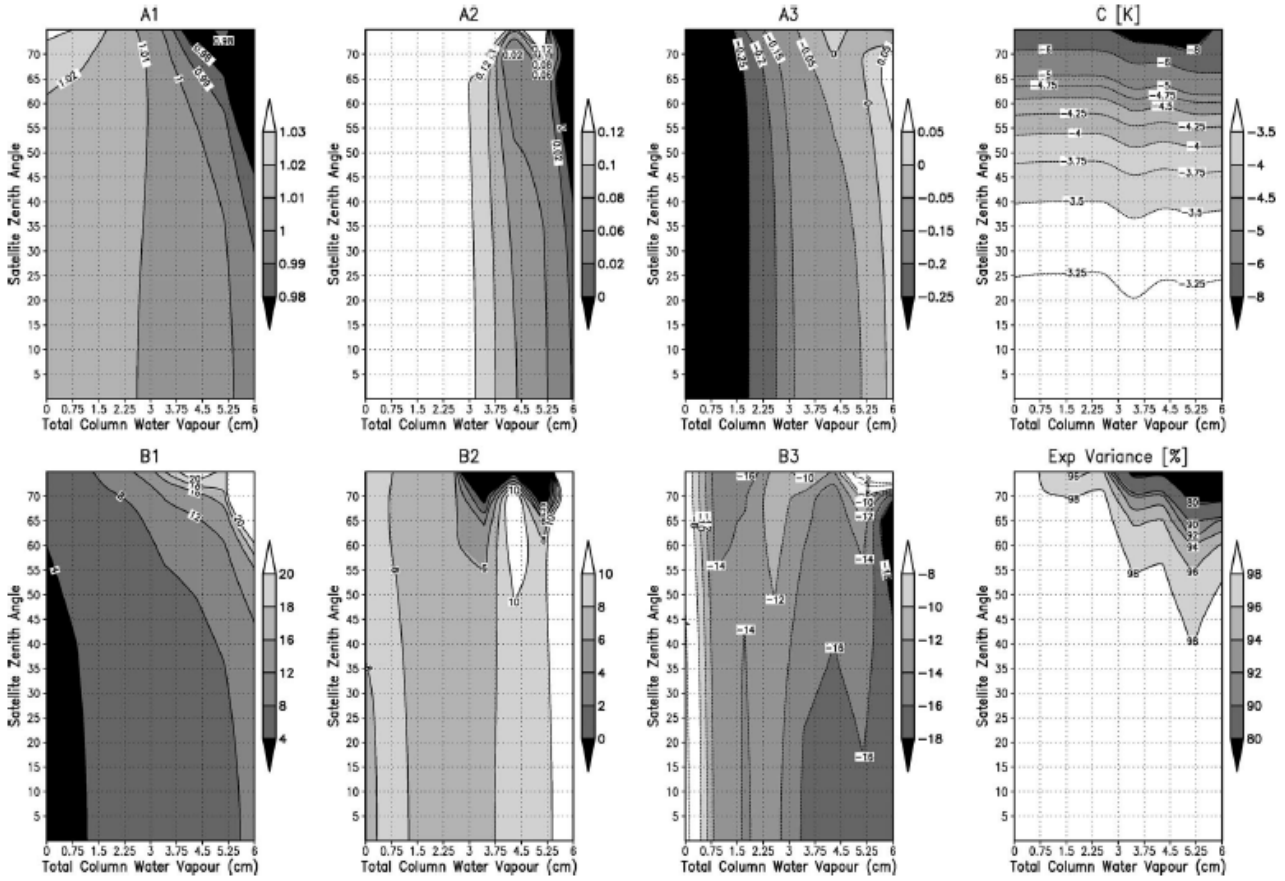


Figure 7: Distribution of the GSW parameters used in SEVIRI (indicated at the top of each panel) and explained variance of the fitted regression (bottom left) as a function of the satellite zenith angle and total column water vapour (in cm) (from Freitas *et al.*, 2010).

Figure 8 shows the SEVIRI GSW error distribution within each class of TCWV and SZA (figure from Freitas et al, 2010). Classes with root mean square differences (RMSD) higher than 4 K are omitted. These classes correspond to cases where explained variance of the SEVIRI GSW within the training dataset is less than 93%, and where errors of 10K or more are commonly obtained within the verification database. Thus, we limit the operational production of LST to SZA below 67.5º when TCWV is 3 cm or higher, and to SZA below 62.5º when TCWV is 4.5 cm or higher.

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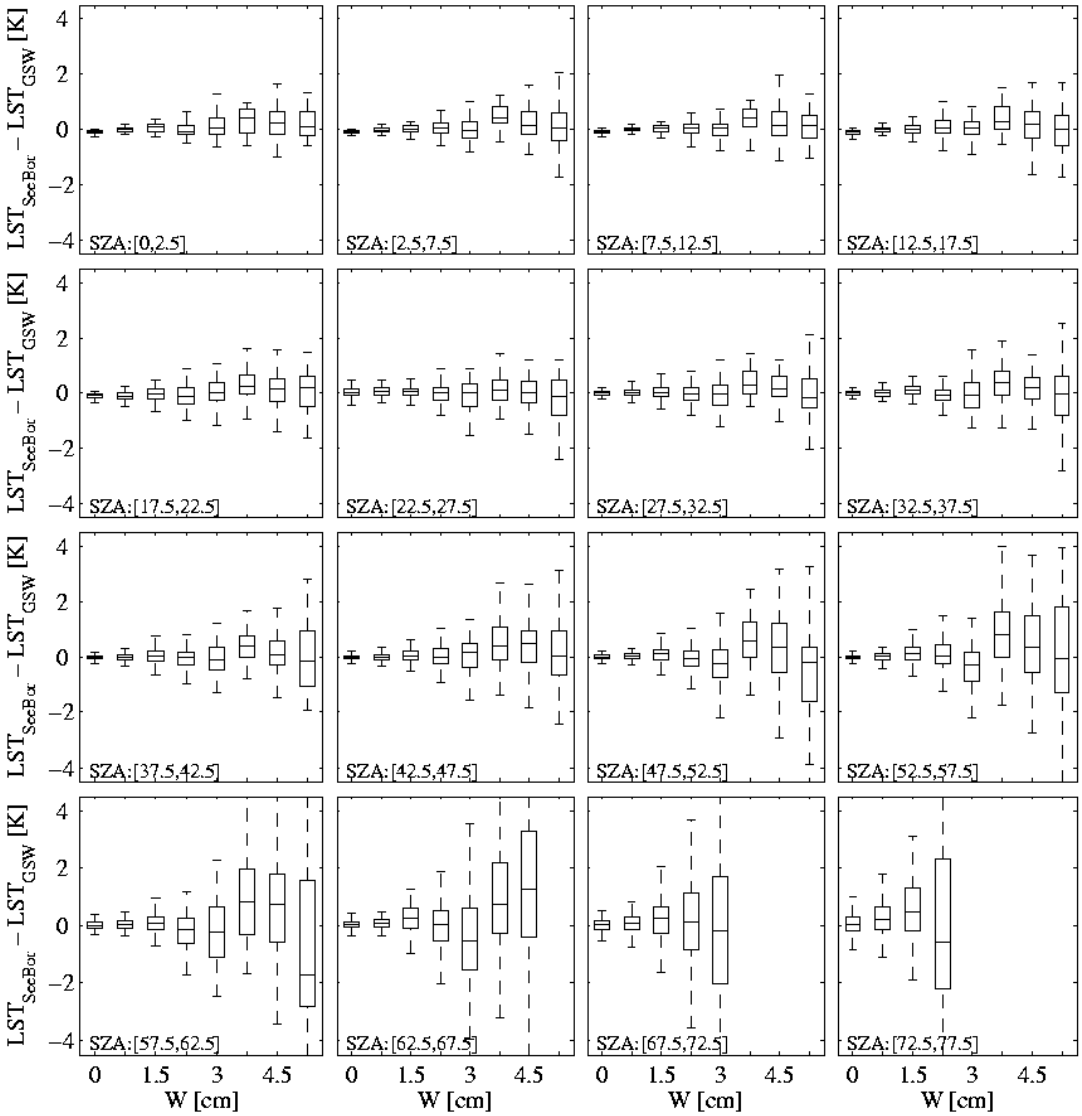


Figure 8: Distribution of LST errors obtained for the SEVIRI GSW verification database which are obtained for different classes of Satellite Zenith Angle (indicated in the bottom left of each panel) and water vapour content (x-axis in each diagram). The lines within each boxplot correspond to the lower quartile, median and upper quartile, respectively, while the whiskers extend to remaining data (from Freitas *et al.*, 2010).

The overall bias and RMSD of the SEVIRI GSW are 0.05 K and 0.78 K, respectively. As shown in Figure 8, the retrieval errors tend to increase with both satellite zenith angle and TCWV. The RMSD is always below 2 K for water vapour content and angles within the range of values required for LST estimations, with the exception of (i) TCWV above 5.25 cm and satellite zenith angle higher than 57.5º; and (ii) TCWV above 2.25 cm and satellite zenith angle higher than 72.5° where the GSW presents RMSD of the order of 3 K.

In the case of AHI and ABI, the model coefficients (Eq.(1)) were updated in product version 2.0. Ten different calibration / validation databases (of radiative transfer simulations) were computed for each sensor. Each database used a different set of atmospheric profiles obeying the distribution described above (i.e., filling the TCWV/SZA "phase space"), chosen randomly. The adopted set (illustrated in Figure 6, common to both AHI and ABI) was chosen for being the one combining a relatively smooth distribution of model coefficients with good validation statistics in the TCWV/SZA space (since the overall performance is reasonably similar among databases). This distribution is shown in Figure 9. Since the channel response functions of ABI and AHI are fairly similar and the set of profiles used for calibration are the same, the coefficients for AHI are also very similar to those of ABI and therefore are not shown here for the sake of brevity. This recalibration allows similar performances for both instruments in v2.0.

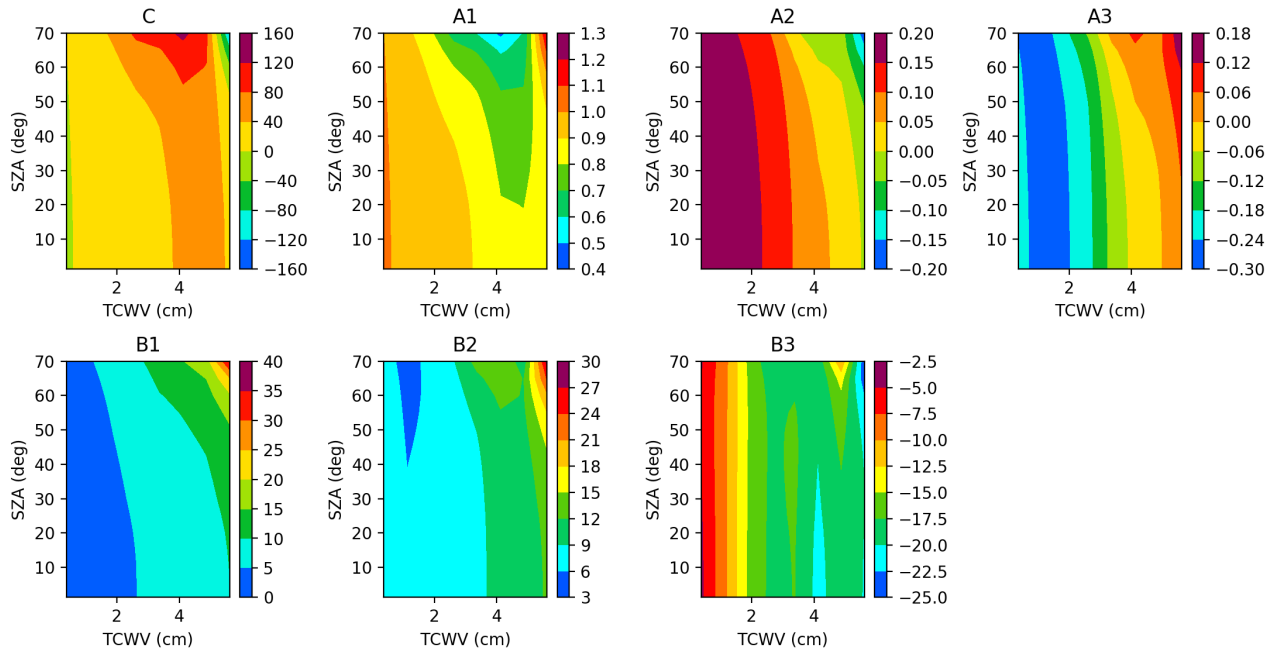


Figure 9: Distribution of the GSW parameters used in ABI/GOES16 (indicated at the top of each panel) as a function of TCWV (in cm) and SZA.

Figure 10 presents the RMSD of the GOES-16, MSG/SEVIRI and Himawari GSW as a function of satellite zenith angle and TCWV classes. Differences between the GSW algorithm developed to the different sensors arise from distinct channel response functions, instrument noise, and differences on the calibration database. Particularly in the case of MSG, we use LSA-SAF LST directly, which used a less comprehensive calibration database. For GOES-16 and Himawari, the calibration database was expanded using the strategy described in Martins *et al*. (2016).

Although some key steps were taken to increase the harmonization of the product, a more harmonized treatment for all sensors is envisaged in future versions. Namely, with the introduction of MTG, all sensors will be calibrated using the same strategy.

The GSW flow chart is shown in Figure 11.

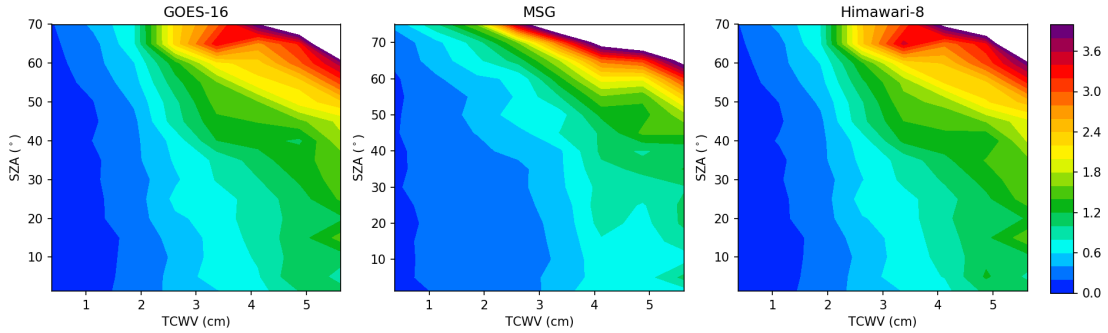


Figure 10: Model RMSD (in K) for the GSW developed for GOES-16 (left), SEVIRI (center) and Himawari-8 (right) thermal infrared window channels.

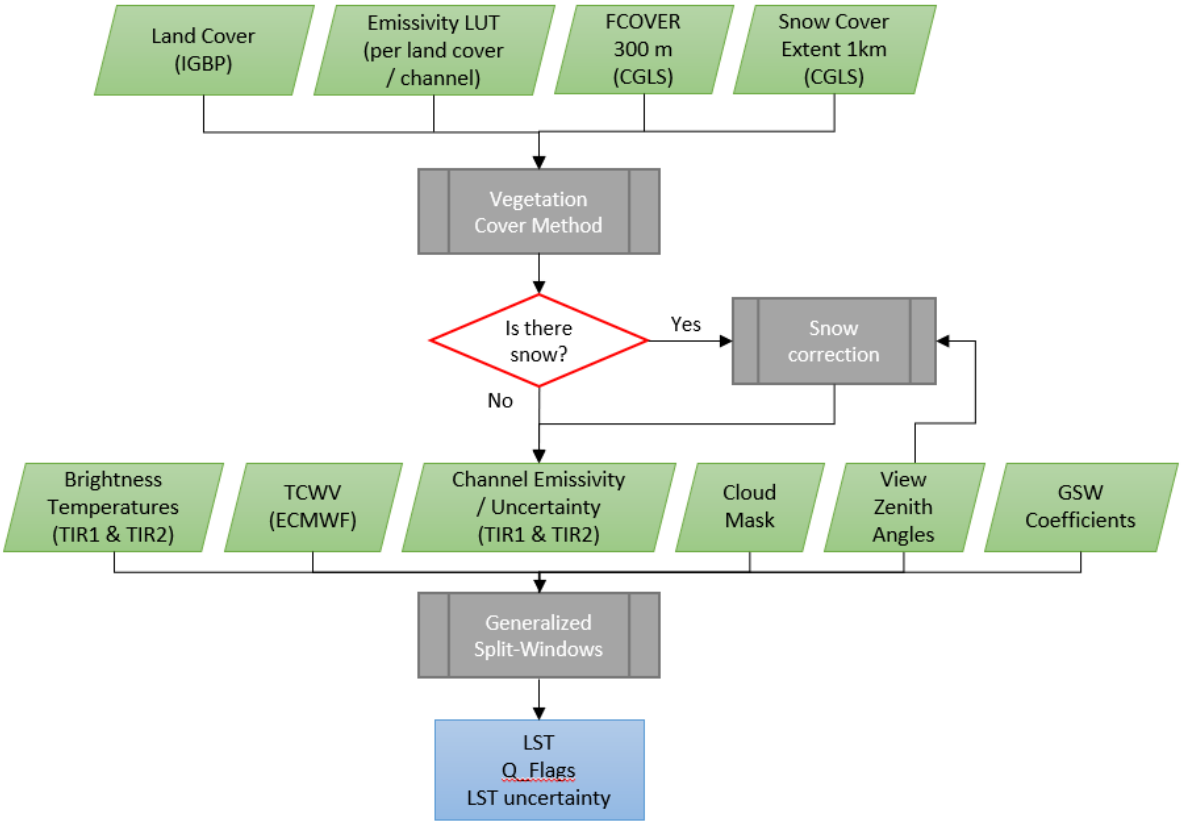


Figure 11: Flow diagram of the generation of LST, its Quality Flags (Q_FLAGS) and associated uncertainty, using the GSW algorithm. Channel emissivity is estimated using the Vegetation Cover Method.

3.2.5.3 LST retrieval uncertainty

The estimation of the uncertainty for each LST value is based on the framework developed by Freitas et al. (2010) for the case of the LSA-SAF LST product. The method relies on clear identification of the uncertainty sources associated to inputs and to the regression model, and their propagation to the final product, which is detailed below. Surface emissivity, atmospheric total column water vapor, the viewing geometry and, to a lesser extent, the sensor radiometric noise are the main factors affecting the uncertainty in the LST retrieval. Barren soils and sparsely vegetated surfaces are characterized by relatively low emissivities values and high uncertainties, since the variability of emissivity among pixels within these land cover types is not well captured in the maps used in LST retrievals. Emissivity errors are higher under dry atmospheres which further contribute to IR retrievals being particularly challenging over arid and semi-arid regions. The presence or absence of snow introduces another source of uncertainty in the retrieval. However, it can be considered as part of the uncertainty due to land cover classification, which is not treated explicitly in the current framework, but is generally captured in the uncertainty due to surface emissivity. Snow and ice have important directional and spectral effects on surface emissivity, whose magnitude depends mainly on snow granularity or viewing geometry. Furthermore, for the "Snow / Ice" land cover class, emissivity values for the TIR2 channel show considerably higher variability when compared to TIR1 (see Table 12 to Table 14 in Annex 1: Emissivities and their uncertainties), which may also affect the retrieval quality. This type of uncertainty could be minimized if there was a robust way to distinguish snow and ice surfaces (Snyder *et al.*, 1998). Another important aspect to consider is the uncertainty due to the time mismatch between the last snow cover extent observation and the LST retrieval: the CLMS Snow Cover Extent is a daily product and therefore provides the snow cover from the previous day. This means that snow might have fallen over a given area while the snow mask describes the pixel as snow-free. Or that snow might have melted while the snow mask still classifies the surface as snow-covered. It is worth mentioning that snow occurs on periods of high cloudiness, so LST retrievals are rarely possible on such occasions, and therefore the impacts of the abovementioned situations are likely to be small.

Furthermore, high atmospheric moisture content enhances the non-linear effects in atmospheric absorption/emission of the IR radiation and increases the optical path, limiting the retrieval by a linear regression scheme such as the GSW (in these cases, the model regression uncertainty is also high – see Figure 10). The optical path is also increased for high satellite viewing angles, and therefore the uncertainty generally increases towards the satellite disk edge.

Finally, errors in the cloud masking algorithm may severely affect the LST retrieval and are usually characterized as negative outliers when compared to in situ values in mostly any product validation exercise. According to validation results of the NWC SAF cloud mask for SEVIRI, the expected rate of missed clouds is of the order of 4% [NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0]. These missed cases often correspond to broken clouds or cases in neighbouring cloudy pixels. It is very difficult to propagate the uncertainty in cloud identification to LST error bars (Bulgin *et al.*, 2018).

In a real scenario, we do not have access to the exact GSW inputs $X = (T_{10.5}, T_{12.3}, \epsilon_{10.5}, \epsilon_{12.3})$ and $Y = (W, \Psi)$ (where W denotes the TCWV and Ψ is the viewing zenith angle), but only to inaccurate inputs, which we denote by $\hat{X} = \left(\hat{T}_{10.5}, \hat{T}_{12.3}, \hat{\epsilon}_{10.5}, \hat{\epsilon}_{12.3}\right)$ and $\hat{Y} = (\widehat{W}, \widehat{\Psi})$. Therefore, if we still infer the LST according to model (1) replacing the exact GSW inputs with the inaccurate ones, we have a new source of uncertainty on the top of the fitting error ΔLST. In the current section, the main uncertainty sources are identified and their impact on the total LST uncertainty is estimated.

Let us define the vector of model coefficients $\theta = (A_1, A_2, A_3, B_1, B_2, B_3, C)$. Notice that the vector θ generated by the fitting process is a function of TCWV and viewing zenith angle i.e., (Y) . Consider the LST estimator $\widehat{LST} = f(\hat{X}, \hat{\theta})$, where $\hat{\theta} = \theta(\hat{Y})$ and $f(X, \theta)$ is the LST estimate given by model (1). A characterization of the model uncertainty is given by:

$$\boldsymbol{S_{LST} = E\left[\left(f(\hat{X}, \hat{\theta}) - LST\right)^2 \middle| X, Y\right]^{1/2}} \tag{9}$$

where the operator $E[.|X, Y]$ stands for mean value conditioned to X and Y; i.e. for a given GSW input X, Y, we want to compute the RMSD of the LST estimate. Using the fact that $LST = f(X, \theta) + \Delta LST$ and assuming that $E\left[f\left(\hat{X}, \hat{\theta}\right)|X, Y\right] = f(X, \theta)$, we may write:

$$\boldsymbol{S_{LST}^2 = E\left[\left(f(\hat{X}, \hat{\theta}) - f(X, \theta)\right)^2 \middle| X, Y\right]^{1/2} + \Delta LST^2} \tag{10}$$

By taking a linear approximation of $f\left(\hat{X}, \hat{\theta}\right)$ in the neighbourhood of (X, θ) and denoting $\sigma_{X_i}^2 = E\left[\left(\hat{X}_i - X_i\right)^2 | X\right]$ and $\sigma_{\theta_i}^2 = E\left[\left(\hat{\theta}_i - \theta_i\right)^2 | Y\right]$, we are led to:

$$\boldsymbol{S_{LST}^2 = \sum_i \left(\frac{\partial f}{\partial X_i}\right)^2 \sigma_{X_i}^2 + \sum_j \left(\frac{\partial f}{\partial \theta_j}\right)^2 \sigma_{\theta_i}^2 + \Delta LST^2} \tag{11}$$

where we have assumed that the components of X, Y are mutually independent and that the $E\left[\left(\hat{X}_i - X_i\right)| X\right] = 0$ and $E\left[\left(\hat{\theta}_i - \theta_i\right)| Y\right] = 0$. Next, we study in detail the error due to each individual GSW input.

3.2.5.3.1 Impact of sensor noise on LST uncertainty

The impact of sensor noise on LST is given by the sum of the uncertainties associated to the noise of each split-window channel:

$$S_T^2 = S_{T_{IR1}}^2 + S_{T_{IR2}}^2 \quad (12)$$

where:

$$S_{T_{IR1}}^2 = \left(\frac{\partial f}{\partial T_{IR1}}\right)^2 \sigma_{T_{IR1}}^2 \text{ and } S_{T_{IR2}}^2 = \left(\frac{\partial f}{\partial T_{IR2}}\right)^2 \sigma_{T_{IR2}}^2 \quad (13)$$

In this case, the value of $\sqrt{\sigma_{T_{IRn}}^2}$ is given by the instrument radiometric noise, $NE\Delta T$, to be reported for each channel upon instrument calibration. In the case of the currently operational sensors in use for the global CLMS LST, all channels / instruments have a reported $NE\Delta T = 0.1$K.

In the case of the GSW algorithm [Eq.(1)], the derivatives with respect to the split-window brightness temperatures are given by:

$$\frac{\partial f}{\partial T_{IR1}} = \frac{1}{2}\left[A_1 + B_1 + (A_2 + B_2)\frac{1-\epsilon}{\epsilon} + (A_3 + B_3)\frac{\Delta\epsilon}{\epsilon^2}\right] \quad (14)$$

$$\frac{\partial f}{\partial T_{IR2}} = \frac{1}{2}\left[A_1 - B_1 + (A_2 - B_2)\frac{1-\epsilon}{\epsilon} + (A_3 - B_3)\frac{\Delta\epsilon}{\epsilon^2}\right] \quad (15)$$

The expected impact of this uncertainty is assessed here by perturbing each of the cases in the validation database with 200 random normally distributed perturbations to each T_{IR_i} and then computing the variance of the differences between the perturbed vs. unperturbed inputs. The result is shown for each sensor, per class of TCWV and satellite zenith angle, in Figure 12. GOES-16 and Himawari-8 show similar behaviour, with impacts growing towards higher optical paths. MSG shows a slightly higher impact towards higher optical paths, compared to GOES-16 and Himawari-8.

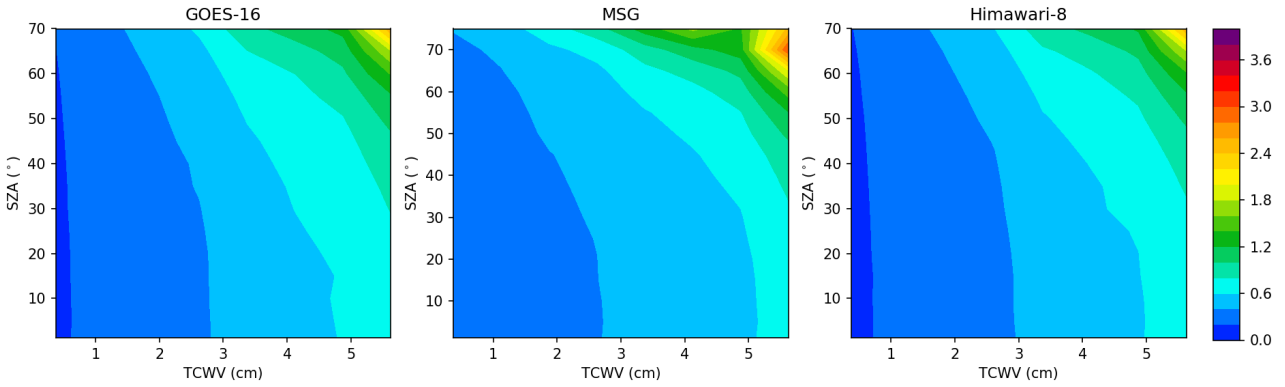


Figure 12: Variability of LST (in K) for GOES-16 (left), MSG (center) and Himawari-8 (right) due to sensor noise, S_T, as a function of TCWV and satellite zenith angle.

3.2.5.3.2 Impact of surface emissivity uncertainties

The impact of uncertainties in surface emissivity for both split-window channels is given by:

$$S_{\epsilon}^2 = S_{\epsilon_{IR1}}^2 + S_{\epsilon_{IR2}}^2 \quad (16)$$

Where

$$S_{\epsilon_{IR1}}^2 = \left(\frac{\partial f}{\partial \epsilon_{IR1}}\right)^2 \sigma_{\epsilon_{IR1}}^2 \text{ and } S_{\epsilon_{IR2}}^2 = \left(\frac{\partial f}{\partial \epsilon_{IR2}}\right)^2 \sigma_{\epsilon_{IR2}}^2 \quad (17)$$

where $\sigma_{\epsilon_{IR1}}$ and $\sigma_{\epsilon_{IR2}}$ are the uncertainty of the respective channel emissivities. To estimate $\sigma_{\epsilon_i}^2$, we need to take into account the uncertainties due to ϵ_{veg_i}, ϵ_{bs_i} and FVC used in Eq. (2). Thus, by propagating their uncertainties using (2), we get:

$$\sigma_{\epsilon_{IR_i}}^2 = FVC^2 \sigma_{\epsilon_{veg_i}}^2 + (1 - FVC)^2 \sigma_{\epsilon_{bs_i}}^2 + \left(\epsilon_{veg_i} - \epsilon_{bs_i}\right)^2 \sigma_{FVC}^2 + \left(d\epsilon_{MR_{IR_i}}\right)^2 \quad (18)$$

where $d\epsilon_{MR_{IR_i}}^2$ is the uncertainty due to multiple reflections. This factor is added here as a source of uncertainty due to the difficulty to explicitly model this phenomenon, which would depend on geometrical factors such as the canopy height and typical separation between vegetation elements (Trigo *et al.*, 2008). Its effect on the emissivity uncertainty for each channel is estimated using (Trigo *et al.*, 2008):

$$d\epsilon_{MR_{IR_i}} = 4\langle d\epsilon \rangle_{IR_i} FVC(1 - FVC) \quad (19)$$

Tabulated values of $\langle d\epsilon \rangle_{IR_i}$ per IGBP land cover type and per channel were calculated in Peres and DaCamara (2005) and are reproduced in Table 8.

Table 8: Values used for $\langle d\epsilon \rangle$ for each split-window channel (see equation (19)). For land covers that are not in the table, values are zero.

IGBP Class	TIR1	TIR2
1; 2	0.0139	0.0123
3; 4	0.0138	0.0122
5	0.0139	0.0122
6	0.0102	0.0088
7	0.0022	0.0019
8	0.0101	0.0087
9	0.0060	0.0052
10	0.0012	0.0011
12	0.0059	0.0048

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13	0.0016	0.0011
14	0.0070	0.0060
16	0.0020	0.0013

To estimate $\sigma^2_{\epsilon_i}$, we need to take into account the uncertainties due to ϵ_{veg_i}, ϵ_{bs_i} and FVC.Thus, by propagating their uncertainties using (2), we get:

$$\boldsymbol{\sigma^2_{\epsilon_{IR_i}} = FVC^2\sigma^2_{\epsilon_{veg_i}} + (1-FVC)^2\sigma^2_{\epsilon_{bs_i}} + \left(\epsilon_{veg_i} - \epsilon_{bs_i}\right)^2\sigma^2_{FVC} + \left(d\epsilon_{MR_{IR_i}}\right)^2} \quad (20)$$

In the presence of snow, that source of uncertainty needs to be considered. This extra term is computed for pixels with snow/ice land cover and for pixels where some snow is detected (i.e., SCE>0.). The CLMS Snow Cover Extent QAR [CGLOPS2_QAR_SCE-NHEMI-1km_V1] reports a mean unbiased RMSD of 0.13, therefore this value is assumed for σ_{snow}. Assuming zero uncertainty due to SZA, and based in Eq. (3), the uncertainty due to snow is estimated as:

$$\boldsymbol{\sigma^2_{\epsilon_{IR_i}} = SCE^2\sigma^2_{\epsilon_{snow_i}} + (1-SCE)^2\sigma^2_{\epsilon_{nosnow_i}} + \left(\epsilon_{snow_i} - \epsilon_{nosnow_i}\right)^2\sigma^2_{snow},} \quad (21)$$

where $\sigma^2_{\epsilon_{snow_i}}$ is estimated considering the variability of snow/ice spectra for the materials in the spectral database and the viewing angle (see values in Table 15), and $\boldsymbol{\sigma^2_{\epsilon_{nosnow_i}}}$ is $\boldsymbol{\sigma^2_{\epsilon_{IR_i}}}$ from Eq. (20). Finally, the derivatives in (17) are given by:

$$\frac{\partial f}{\partial \epsilon_{IR1}} = \frac{1}{2}\left[-\frac{A_2}{2}\frac{T_{IR1}+T_{IR2}}{\epsilon^2} + A_3\left(T_{IR1}+T_{IR2}\right)\left(\frac{\epsilon-\Delta\epsilon}{\epsilon^3}\right) - B_2\frac{T_{IR1}-T_{IR2}}{\epsilon^2} + B_3\left(T_{IR1}-T_{IR2}\right)\left(\frac{\epsilon-\Delta\epsilon}{\epsilon^3}\right)\right] \quad (22)$$

$$\frac{\partial f}{\partial \epsilon_{IR2}} = -\frac{1}{2}\left[\frac{A_2}{2}\frac{T_{IR1}+T_{IR2}}{\epsilon^2} + A_3\left(T_{IR1}+T_{IR2}\right)\left(\frac{\epsilon+\Delta\epsilon}{\epsilon^3}\right) + B_2\frac{T_{IR1}-T_{IR2}}{\epsilon^2} + B_3\left(T_{IR1}-T_{IR2}\right)\left(\frac{\epsilon+\Delta\epsilon}{\epsilon^3}\right)\right] \quad (23)$$

In Figure 13, an assessment of this source of uncertainty is shown as a function of TCWV and for 3 classes of mean emissivity (between TIR1 and TIR2): arid ($\epsilon \leq 0.96$), sparsely vegetated ($0.96 < \epsilon \leq 0.98$) and highly vegetated / water ($\epsilon > 0.98$). For each profile of the validation database, a reference retrieval was made using the FVC, ϵ_{bs} and ϵ_{veg} values of Table 12 to Table 14, corresponding to the land cover assigned to the profile. Then, 200 random normally distributed perturbations of FVC were introduced in each case of the validation database with a standard deviation equal to the reported uncertainty in FVC, as well as 200 uniformly distributed perturbations of ϵ_{bs} and ϵ_{veg} with distribution half-widths corresponding to the respective uncertainty. Results for GOES-16 and Himawari-8 are very similar, with 1) a general decrease of S_ϵ with TCWV due to compensation of errors between emitted land surface radiance and reflected radiance emitted by the atmosphere (which increases in moister atmospheres) and 2) with larger values in situations with less vegetation, which are characterized by lower and more uncertain emissivities. MSG shows somewhat higher values (with the exception of emissivity values characteristic of sparsely vegetated surfaces), not only due to differences in the calibration database but also due to differences in the channel

frequencies/response functions of each channel. Bare ground emissivity uncertainties are higher for sparsely vegetated land covers (e.g., savannas) but are relatively lower at the 10.8 µm channel used by MSG when compared to the 11.2 µm channel used by both ABI and AHI). The same is true for the TIR2 channels (see tables in Annex 1: Emissivities and their uncertainties).

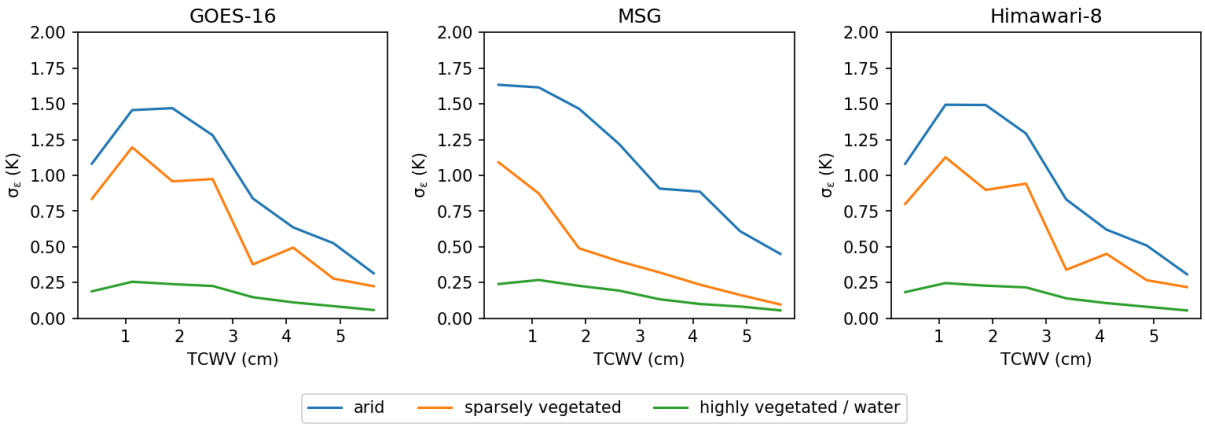


Figure 13: Variability of LST for GOES-16 (left), MSG (center) and Himawari-8 (right) due to emissivity uncertainty, S_{ϵ}^{2}, as a function of TCWV. Values are given for 3 classes of mean emissivity (between TIR1 and TIR2): arid ($\epsilon \leq 0.96$) in blue, sparsely vegetated ($0.96 < \epsilon \leq 0.98$) in yellow and highly vegetated / water ($\epsilon > 0.98$) in green.

3.2.5.3.3 Uncertainties in forecasts of atmospheric water vapour content

Since the total column water vapor is used implicitly in the algorithm (i.e., different sets of parameters are used for each class of TCWV values), the uncertainty cannot be calculated analytically. The uncertainty due to this parameter is related to the fact that there is a probability of choosing the wrong set of model coefficients because the TCWV forecast may lead to a wrong choice of TCWV class. Therefore, it is estimated *a priori* as follows:

1) the operational use of the GSW algorithm (Eq.1) to retrieve LST from each sensor makes use of forecasts of TCWV (W) provided by the European Centre for Medium-range Weather Forecasts (ECMWF). To characterize W error statistics, we compare ECMWF W forecasts (with forecast steps ranging between 12 and 36 h) with the respective analysis, for the 15th of each month for one full year; ECMWF grid points with model cloud cover higher than 10% were excluded. This exercise is repeated regularly (about once per year) to update the uncertainty in ECMWF forecasts. The comparison between W forecasts and analysis (the reference value) allows us to estimate the probability $P\left(\widehat{W}_{i_{fc}}|W_{i_{an}}\right)$, i.e., the probability that $\widehat{W}$ belongs to the water vapour content class $W_{i_{fc}}$, given that the true class is $W_{i_{an}}$.

2) The validation database is used to calculate the variance of the LST for each class of SZA/TCWV. The variance is obtained by simulating each case of the validation database (i_{valid}) with all the possible sets of coefficients corresponding to that SZA (i_{fc}) and then comparing to the value obtained with the "true" set of coefficients ($LST_{i_{valid}}$). The uncertainty final estimate is provided for each SZA / forecast TCWV class by multiplying the obtained variance by the probability of a given forecast class does not correspond to the "true" class:

$$S_W^2 = \frac{1}{n_{TCWV}} \sum_i^{n_{TCWV}} \sigma^2_{\widehat{W}_{i_{an}}|W_{fc}} P(W_{i_{an}}|\widehat{W}_{fc}) \quad (24)$$

where $\sigma^2_{\widehat{W}_{i_{an}}|W_{fc}}$ is the variance of the errors of LST for a given W estimate (i.e., for a given TCWV forecast $\widehat{W}_{fc}$), when the correct value is $W_{i_{an}}$; $P(W_{i_{an}}|\widehat{W}_{fc})$ is the probability that the correct class of TCWV is determined by $W_{i_{an}}$, conditioned by the forecast value of $\widehat{W}_{fc}$; n_{TCWV} is the number of TCWV classes considered.

In Figure 14, the variability due to uncertainty in LST is shown as function of TCWV and satellite zenith angle (SZA). Again, higher values occur towards higher optical paths (i.e., towards higher TCWV and higher SZA). In general, the uncertainty due to TCWV is small because the forecasts are generally close to the analysis used for verification (therefore it is unlikely that a set of coefficients corresponding to a different TCWV class is used, especially towards lower TCWV values). Only for those classes with higher TCWV, the fact that a different set of coefficients is used will imply higher differences in the retrieved LST.

These values are prescribed to the operational chain as a LUT, since it is not possible to perform these calculations for each forecast (since the analysis is of course not available at the time of production).

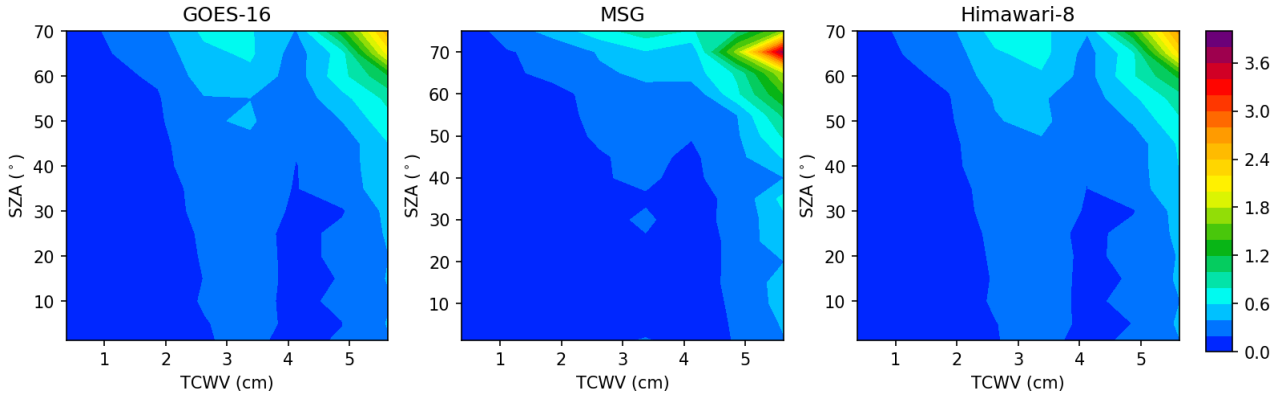


Figure 14: Variability of LST (in K) for GOES-16 (left), MSG (center) and Himawari-8 (right) due to the uncertainty in TCWV forecasts, S_W, as a function of TCWV and satellite zenith angle. White areas denote uncertainties above 4 K.

3.2.5.3.4 Total uncertainty of LST retrievals

The estimation of LST error bars, S_{LST}, assumes that all sources of errors described in the previous sections are independent and therefore is given by:

$$S_{LST} = \sqrt{S_T^2 + S_\epsilon^2 + S_W^2 + \Delta LST^2} \quad (25)$$

In Figure 15, an assessment of the total expected uncertainty is shown as a function of TCWV and satellite zenith angle (SZA), for each sensor and for arid, sparsely vegetated and highly vegetated situations. Overall, uncertainty grows towards higher optical paths, to the point that for a few classes of high TCWV and SZA we do not recommend the usage of the product and therefore all situations with pixel uncertainty higher than 5 K are not taken into account for the final LST product (see section 3.3). The required product target uncertainty is 2 K (see Chapter 2). However, users are advised to take into consideration the estimated error bars for their application. **LST values with uncertainties above 4 K must be used with caution.** These values are usually found in those classes with higher optical path (see white areas in Figure 15).

The analysis of algorithm uncertainties is complemented with validation against independent sources of data, including in situ observations and LST products retrieved from other sensors (Trigo et al, 2008b, Goettsche et al, 2013, 2016, Ermida et al, 2014). As shown in Goettsche et al. (2016), there is good agreement with in situ measurements taken in arid and semi-arid regions. In fact, for comparisons made between SEVIRI LST retrievals and in situ measurements taken at Gobabeb and Farm Heimat in Namibia over the 2009 - 2014 period, RMSD was generally below 1.5 K, with slightly higher values in months with more undetected clouds, while biases varied between -0.5 K and 0.6 K. These ground stations are located over highly homogeneous areas, and therefore the local measurements represent very reasonably those at the satellite pixel scale. The monthly average uncertainties estimated in NRT for SEVIRI/MSG LST, taking into account expected algorithm and input accuracies, lies between 2 K and 3 K. This range is of the same order or even larger than the root mean square differences between the in situ and the satellite measurements (Trigo et al, 2008b; Freitas et al., 2010).

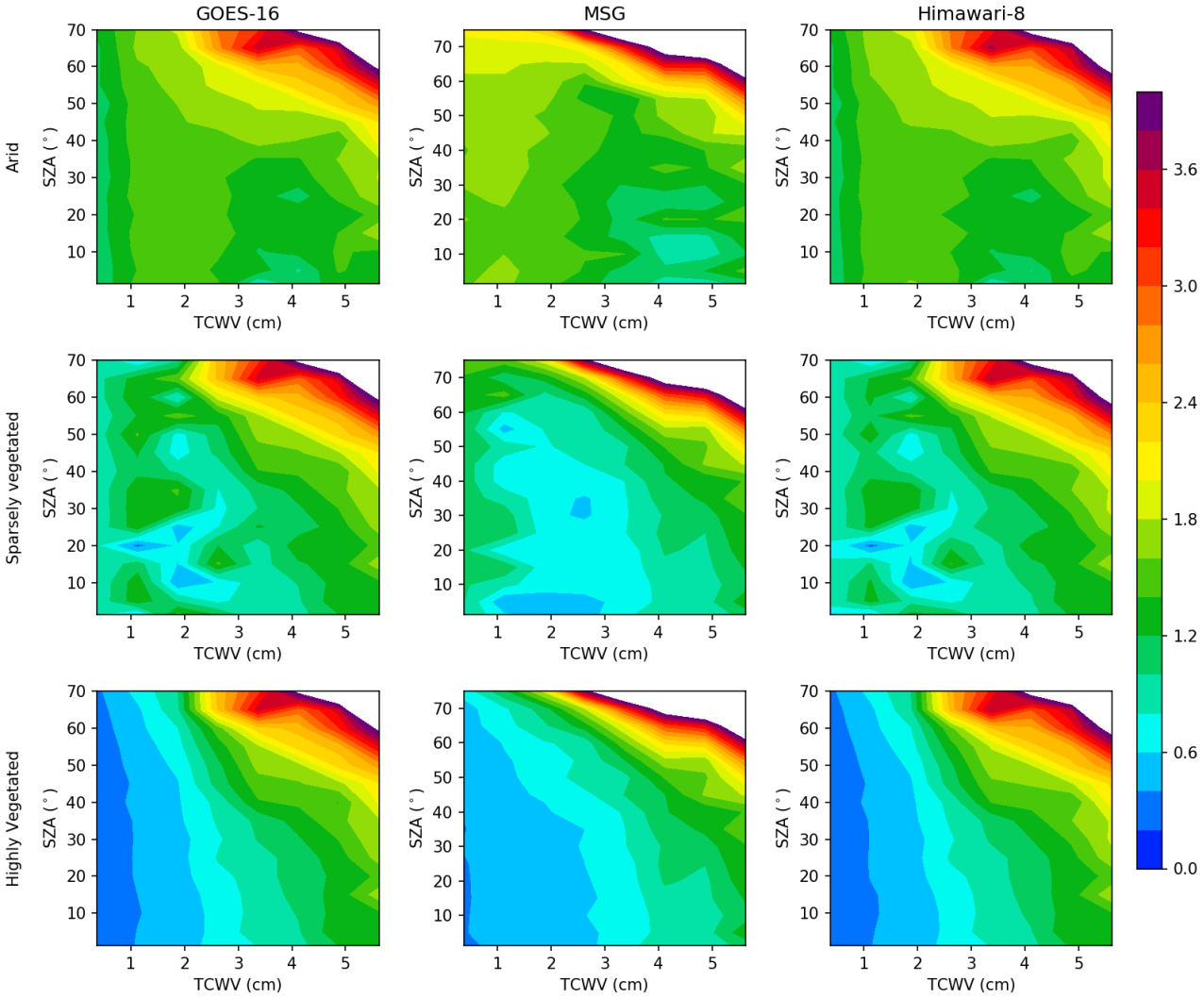


Figure 15: Variability of LST (in K) for GOES-16 (left), MSG (center) and Himawari-8 (right) due to the uncertainty in all inputs, S_{LST}, as a function of TCWV and satellite zenith angle. From top to bottom, results are shown for arid, sparsely vegetated and highly vegetated situations.

3.2.6 Dual-Algorithm

3.2.6.1 Input data

The Dual-Algorithm (DA) provides estimation of LST using measurements from (i) one middle infrared and one thermal window channel, T_{MIR} and T_{TIR1} (Table 5) during night-time and; (ii) a single thermal window channel T_{TIR1} during daytime. The reason for having two different formulations is to avoid uncertainties in LST due to solar contamination of T_{MIR} during daytime.

A version of the DA was adjusted to GOES-13 (and also its predecessors) and MTSAT imagers since the GSW can only be applied whenever the two IR channels are available in NRT.

The DA requires the following inputs:

- Land Cover data base - provided by the International Geosphere-Biosphere Program (IGBP) database (Belward, 1996).
- Cloud mask developed by the project team, based on the NWC SAF software (see Annex 3: Cloud mask).
- Total Column Water Vapour (TCWV), which is obtained from ECMWF operational forecasts, retrieved with an hourly frequency and at 9 km resolution.

The DA does not rely explicitly on emissivity, as the model coefficients are calibrated for classes of TCWV, SZA and Land Cover. The effect of emissivity is dealt implicitly, by determining different coefficients for each land cover class. This method is therefore more uncertain than the GSW since it does not account for different vegetation covers within each class, nor for the respective seasonal variation.

3.2.6.2 Methodology

Following the approach presented in Sun et al. (2004), a mono-channel and a two-channel algorithm were developed for channels available onboard GOES-13 and MTSAT, respectively. The mono- and two-channel methods are based on equations (3) and (4), and unlike the original formulations developed by Sun et al. (2004), the coefficients depend implicitly on TCWV.

The Mono-Channel Algorithm, used during day time is given by:

$$\boldsymbol{LST = A_1 + A_2 T_{TIR1} + \Delta LST} \quad \textbf{(26)}$$

The Two-Channel Algorithm, used in the night time is given by:

$$\boldsymbol{LST = B_1 + B_2 T_{TIR1} + B_3 (T_{TIR1} - T_{MIR}) + \Delta LST} \quad \textbf{(27)}$$

where ΔLST is the model error and A_i and B_i (i=1,2,3) the model coefficients. As in the case of the GSW algorithm, the coefficients in equations (23) and (24) are fit to a calibration dataset for different classes of TCWV, satellite viewing zenith angle (SZA) and, in this case, also to land cover type since these algorithms do not use explicit emissivity as an input. The calibration and verification databases of the DA are built independently for GOES-13 and MTSAT imagers. They follow the same methodology described in the previous section, relying on radiative transfer simulations of TOA brightness temperatures for the middle and thermal infrared channels available onboard each sensor. Therefore, MODTRAN4 simulations obtained for SeeBor database within the spectral ranges of MIR

and TIR bands are convoluted with the sensors response function, as in equation (3). Note that the dual-algorithm was not recalibrated using MODTRAN6, as the GSW algorithm was (see 3.2.5.2), since no reprocessing activity is foreseen. TOA brightness temperatures are obtained via application of the Planck function.

The parameters in the DA algorithm are estimated for each of the 16 land-cover types within IGBP database, for 8 different classes of TCWV, up to 6 cm, and for 16 classes of SZA, up to 75º, ensuring that all ranges of atmospheric attenuation within the thermal infrared are covered.

The parameters of the Mono-Channel Algorithm and of the Two-Channel Algorithm are schematically presented in Figure 16 and Figure 17 respectively, along with the variance of LST explained by the regression, for 4 different landcover types. The uncertainty of the mono-channel (used for daytime) and two-channel (used for night-time) formulations are assessed by comparison with the verification dataset. The distribution of errors is shown in Figure 18 and Figure 19 for GOES and in Figure 20 and Figure 21 for MTSAT formulations. While the two-channel methodology presents errors similar to those obtained for the GSW – frequently within the 2 K range – the use of a single channel is associated to a considerable uncertainty increase.

As before, the estimation of the uncertainty for each LST value is based on the results presented in Figure 18 to Figure 21, along with a series of sensitivity tests to input errors, namely those associated to (i) ECMWF TCWV forecasts and; (ii) emissivity variability within each land cover database. We will also consider 4K as the maximum uncertainty estimation, measured by the RMSD between reference and estimated LST; higher uncertainty values will be masked out.

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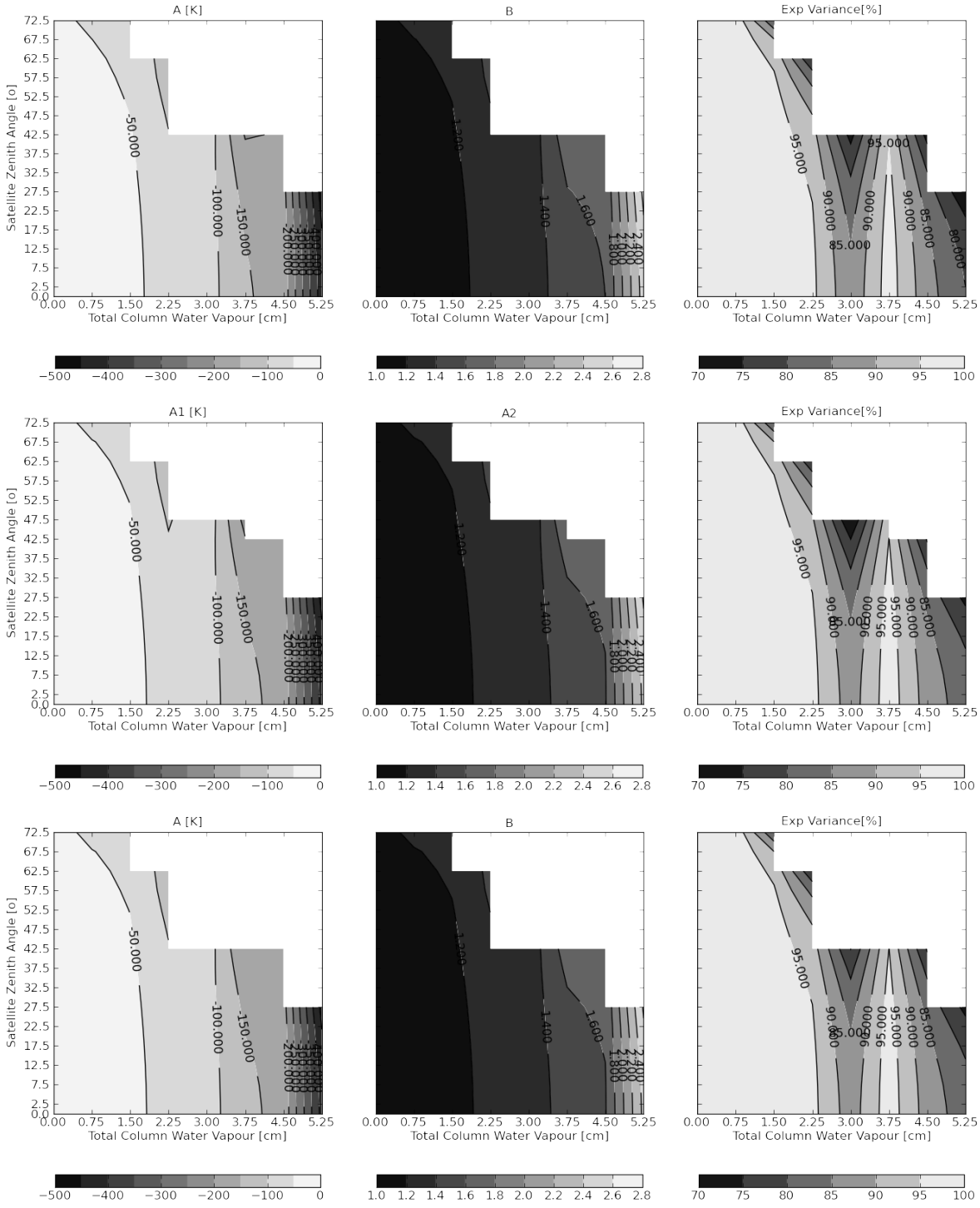


Figure 16: Distribution of the One-Channel Algorithm parameters (indicated at the top of each panel) developed for GOES-13 imager, for different landcover types (from top to bottom Bare Soil, Croplands and Permanent Wetlands) and explained variance of the fitted regression (right) as a function of the satellite zenith angle and total column water vapour (cm). Classes for which explained variance is above 85% and/or the algorithm error exceeds 4 K are masked out.

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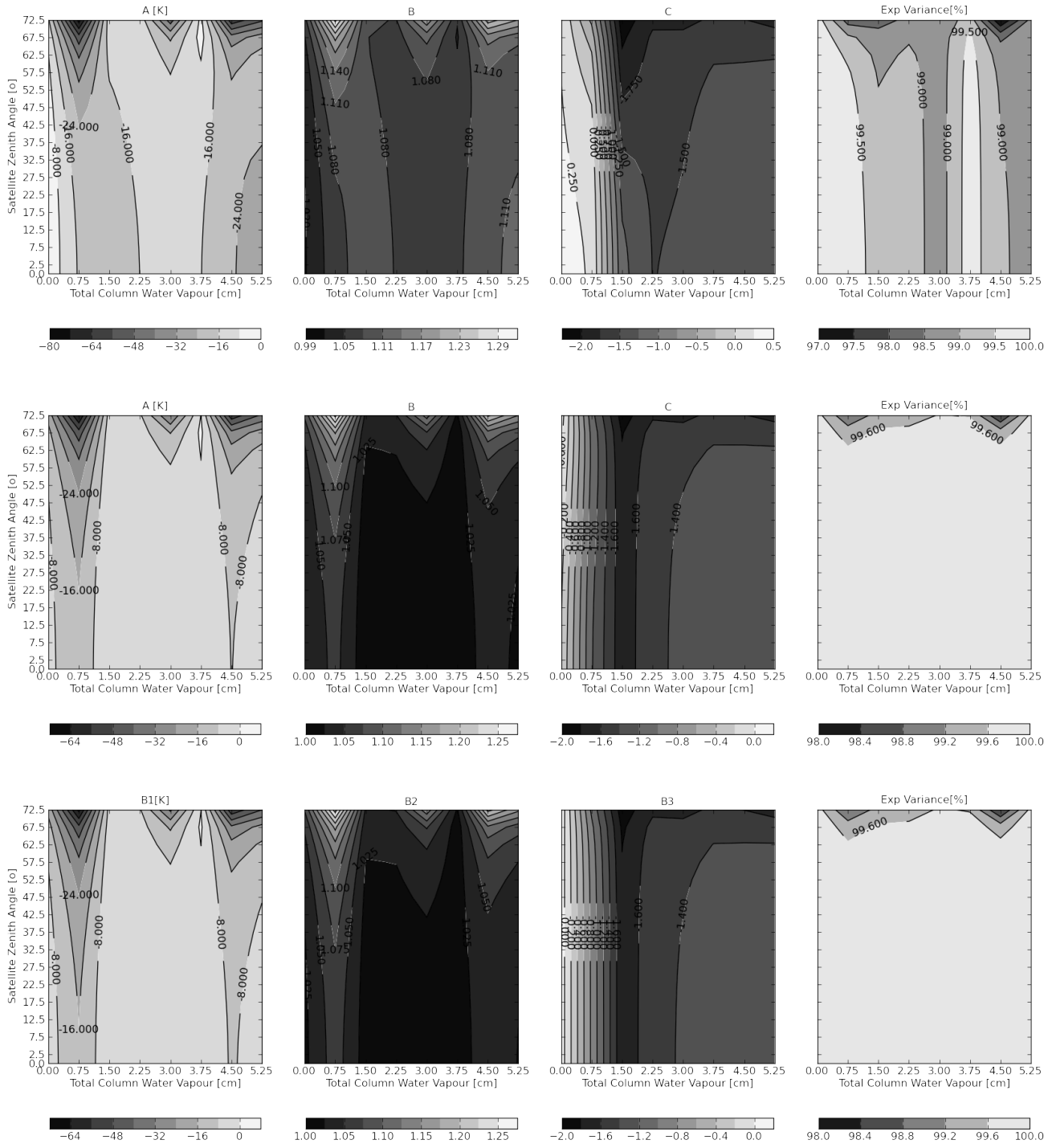


Figure 17: Distribution of the Two-Channel Algorithm parameters (indicated at the top of each panel) developed for GOES-13 imager, for different landcover types (from top to bottom Bare Soil, Croplands and Permanent Wetlands) and explained variance of the fitted regression (right) as a function of the satellite zenith angle and total column water vapour (cm).

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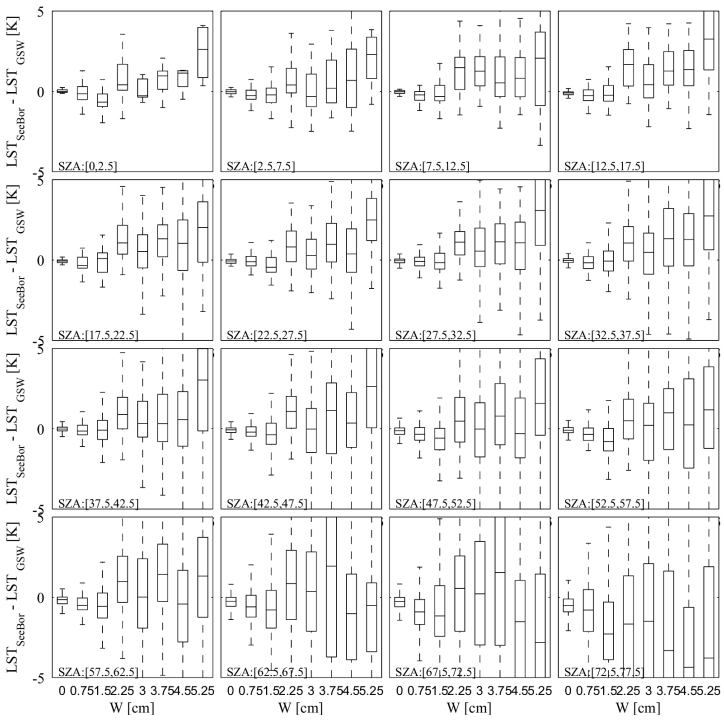


Figure 18: as in Figure 8, but for the mono-channel algorithm developed for GOES-13 window channel.

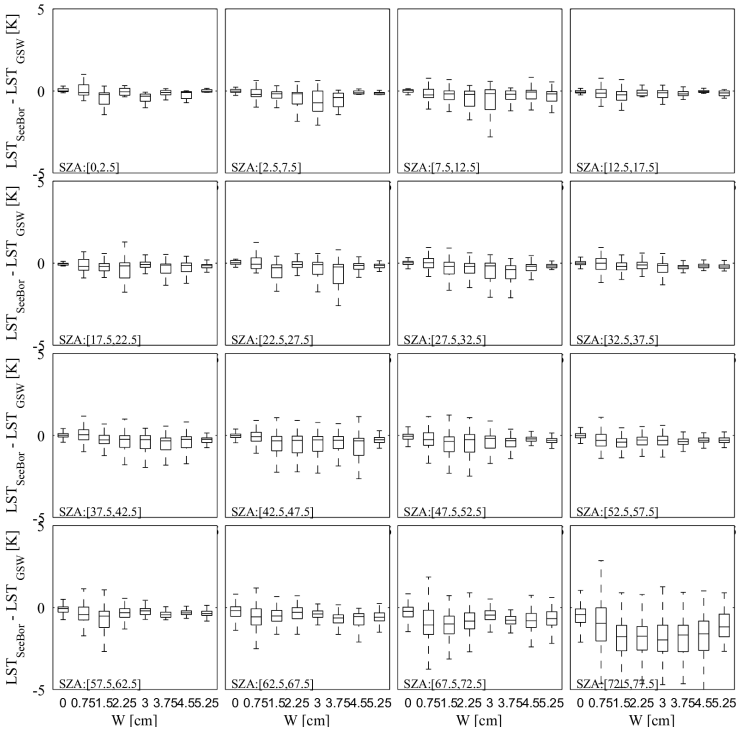


Figure 19: as in Figure 8, but for the two-channel algorithm developed for GOES-13 MIR and TIR channels.

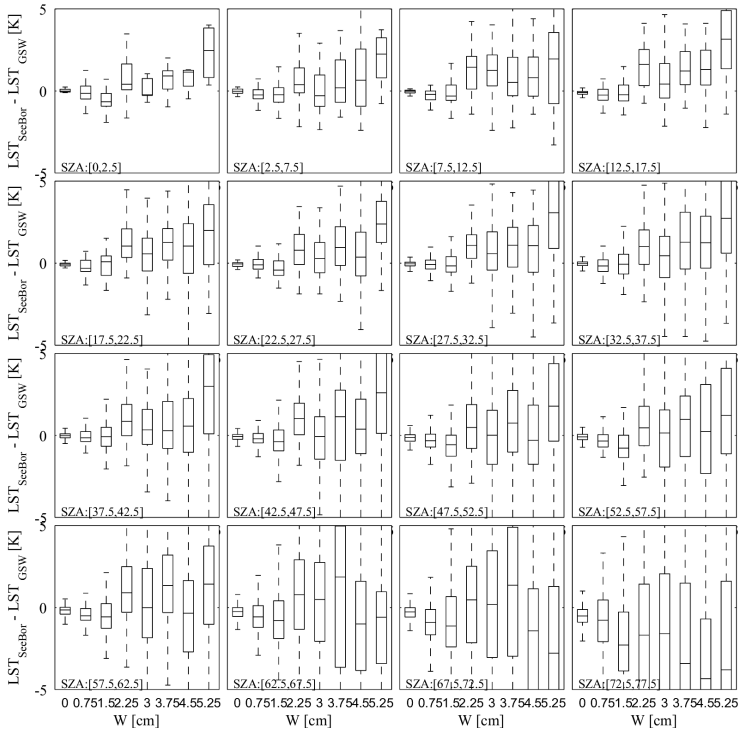


Figure 20: as in Figure 8, but for the mono-channel algorithm developed for MTSAT 11µm channel.

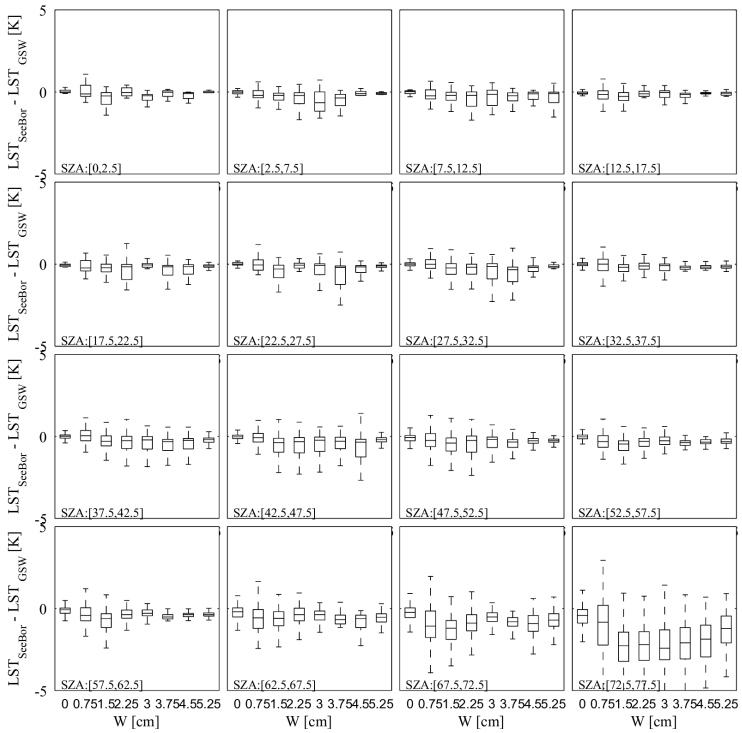


Figure 21: as in Figure 8, but for the two-channel algorithm developed for MTSAT MIR and TIR channels.

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The DA flow chart is shown in Figure 22.

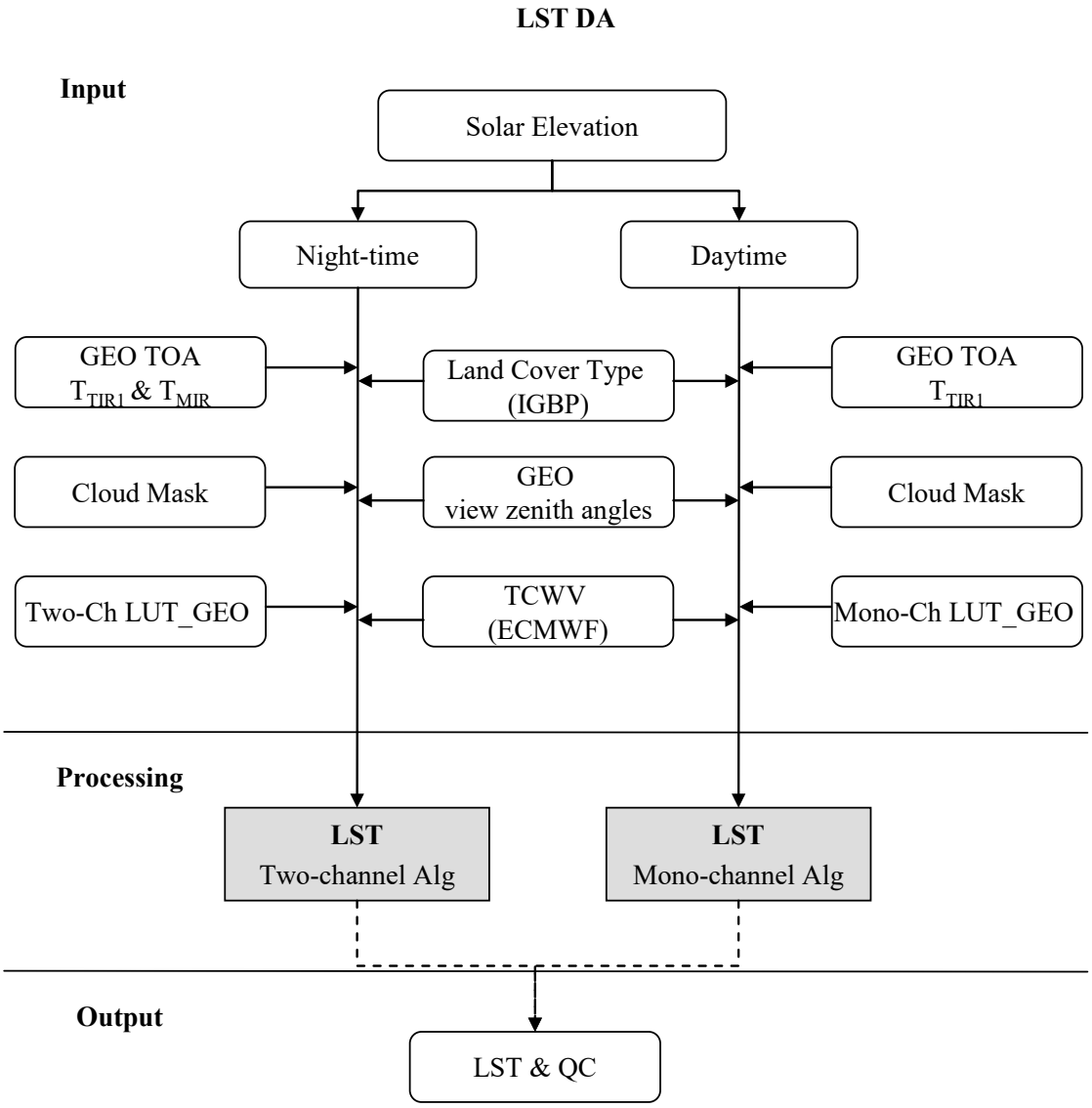


Figure 22: Flow diagram of the generation of LST and respective Quality Control (QC), using the Dual-algorithm. The diagram is valid for both GOES-13 and MTSAT imager data (denoted by GEO).

As already mentioned, the difference between the brightness temperatures of two spectral channels gives information about the atmospheric attenuation in the satellite measurements. In case of the mono-channel algorithm the atmospheric information is only given by the auxiliary input data (TCWV and satellite zenith angle). As such, the mono-channel algorithm is much more dependent on the auxiliary input data than the two-channel algorithms. Besides that the mono-channel algorithm is adapted for specific classes of TCWV and satellite zenith angle, thus a small difference in one of these parameters could imply on a considerable difference in the atmospheric correction which, in some particularly conditions, imply on spatial discontinuities on the LST field. To overcome this

effect an interpolation on the atmospheric correction is applied to the mono-channel derived LST. The interpolation consists in calculate, for each pixel, two values of LST corresponding to two adjacent classes of water vapour. The final LST is a linear interpolation of those values for the actual water vapour content.

3.2.6.3 LST retrieval uncertainty

In the case of DA, a similar framework to the one presented in section 3.2.5.3 is used. The main difference is that the algorithms do not have an explicit emissivity, so we only need to take into account the uncertainty due to land cover (apart from the regression error, sensor noise and TCWV forecasts). For the sake of completeness, we present the formulations used to derive the uncertainty due to sensor noise and due to land cover using DA in the following sub-sections. The treatment of the uncertainty due to TCWV forecasts is similar to what was described in section 3.2.5.3

3.2.6.3.1 Uncertainty due to sensor noise

The reported values of the instrument radiometric noise ($NE\Delta T$) for the channels used in the DA are given in Table 9 for each sensor.

Table 9: Reported $NE\Delta T$ values, corresponding to $\sqrt{\sigma^2_{T_{TIR}}}$ and $\sqrt{\sigma^2_{T_{MIR}}}$ (in K) for the DA channels of each sensor.

	GOES / Imager	MTSAT / JAMI
TIR	0.1281	0.4215
MIR	0.1789	0.3940

The uncertainty due to sensor noise is given by the sum of the contributions of each channel:

$$S^2_T = S^2_{T_{TIR}} \underbrace{\left(+S^2_{T_{MIR}}\right)}_{\text{for two-channel algo}} \tag{28}$$

where:

$$S^2_{T_{TIR}} = \left(\frac{\partial f}{\partial T_{TIR}}\right)^2 \sigma^2_{T_{TIR}} \text{ and } S^2_{T_{MIR}} = \left(\frac{\partial f}{\partial T_{MIR}}\right)^2 \sigma^2_{T_{MIR}} \tag{29}$$

The derivative in Eq. (26) for the mono-channel algorithm is given by:

$$\frac{\partial f}{\partial T_{TIR}} = A_2 \tag{30}$$

In the case of the two-channel, the derivatives are given by:

$$\frac{\partial f}{\partial T_{TIR}} = A_2 + A_3 \tag{31}$$

$$\frac{\partial f}{\partial T_{MIR}} = -A_3 \tag{32}$$

3.2.6.3.2 Uncertainty due to emissivity

To account for emissivity, the two-channel algorithm is calibrated for classes of land cover. Each land cover comprises a range of materials, each with different spectral emissivities for bare ground and vegetation, as well as a typical fraction of vegetation cover. This variability within each land cover class is prescribed to the processing chain as a LUT, which takes into account all cases in the validation database.

3.3 DATA FUSION

All algorithms described in the previous sections allow the estimation of LST on a pixel-by-pixel basis. LST corresponds to instantaneous fields produced every 15 minutes in the case of SEVIRI, and hourly in the case of GOES and Himawari. The three individual fields are then re-projected and merged onto a 5/112ºx5/112º regular grid, according to user requirements. From the version 2.0 onwards, the individual timeslots selected to integrate the global product are T=0 for all sensors. In previous versions (v1.x), the T=0 time slot was used for both GOES and Himawari, while T-15 was used for MSG. When the product was originally designed, this configuration minimized differences between acquisition times within regions where the individual satellite disks overlap. With the introduction of Himawari, the T=0 slot was selected since there was no overlap with any other disk. With the introduction of GOES-16 (and following satellites) in the processing chain (v1.2), the optimised solution (i.e., with lower differences in the acquisition times among sensors) was to use T=0 for all sensors. However, the production team decided to maintain consistency in the time series obtained within the MSG disk and kept T-15 for MSG. Since v2.0 has represented a major upgrade with the inclusion of the IODC mission, it was therefore decided to increase spatial consistency by using T=0 for all sensors since there is an increase of the overlapping areas, even if this introduces a discontinuity in the time-series of pixels within the MSG 0° area. Users must be aware of these changes when performing any analysis using global CLMS LST products. The production team will warn users about reprocessing activities that might occur.

LST data fusion consists of LST retrievals from the different geostationary satellites merged into a single field, thus yielding a global product. The final LST product is obtained by averaging all the LST values from pixels within each grid box (regular in latitude and longitude), for satellite zenith angles up to 70º for all sensors. In this context, all the averages are weighted considering the

respective viewing angle, so that the final product is less influenced by lower quality retrievals (i.e., those with greater satellite zenith angle). At the end, four data layers are added to the LST product with additional information crucial for the product applications: i) the error bars in a pixel by pixel basis, determined taking into account algorithm uncertainties as well as the propagation of input uncertainties (section 3.2; Freitas et al., 2010); ii) the pixel acquisition time – in areas observed by more than one satellite, this value corresponds to an average of acquisition times, also weighted by the satellite zenith angle; iii) the fraction of pixels that were effectively processed, and iv) a quality flag with satellite information, missing data, cloudiness and land-water mask (Table 16 in Annex 2: LST Quality Control Information).

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4 KNOWN LIMITATIONS

The vegetation cover method has been widely used to estimate surface emissivity in the thermal infrared bands used for LST retrievals requiring this information (Becker and Choudhury, 1988; Caselles *et al.*, 1997; Peres and DaCamara, 2005; Trigo *et al.*, 2008). This method assumes that a given bare soil emissivity, ϵ_{bg}, and a given vegetation emissivity, ϵ_{veg} for each land cover class are representative of the whole land cover class. This is not always the case, e.g., in croplands, urban or barren areas, where differences in soil composition, crop/vegetation density or presence of man-made infrastructures may indeed correspond to different surface emissivities within the same land cover class. In the algorithm used here, this emissivity variability within each land cover class is accounted for by prescribing adequate uncertainties for bare soil and vegetation emissivities (σ_{bs} and σ_{veg}, respectively). These values are obtained by estimating the maximum absolute deviation of the emissivities (both ϵ_{bs} and ϵ_{veg}) within each class, using all the materials from the spectral emissivity database assigned to that class. This way, the method provides an upper estimate of the uncertainty due to emissivity variability within each land cover class. The more comprehensive the spectral database is, the more accurate is the uncertainty estimate using this method.

Careful analysis of Table 12, Table 13 and Table 14 shows that the relative uncertainty of emissivities within each class is below 1% except for land covers 15 and 16 (Snow/Ice and Barren), where it may reach 3%. The effect of vegetation cover (i.e due to the differences between ϵ_{bs} and ϵ_{veg} within each class) for all classes is of the order of 3%. Accounting for the effect of vegetation cover was key to reduce the uncertainty due to emissivity in all vegetated land cover classes. Methods to improve the estimation of bare soil emissivity are subject of current research (e.g., Borbas *et al.*, 2018; Hulley *et al.*, 2015; Jimenez-Munoz *et al.*, 2014; Malakar and Hulley, 2016; Masiello *et al.*, 2018) and may be considered for future versions of the global CLMS LST product after they reach an acceptable maturity level. The results in the QAR [CGLOPS1_QAR_LST-LST10-V2.1] show that there is room for improvement in order to reduce the uncertainty for snow/ice and barren pixels. A more sophisticated cloud masking algorithm should be adopted, especially to help disentangling snow from cloud-over-snow situations which are compromising product quality over such scenes.

5 OUTPUT PRODUCT

The LST files contain one main dataset corresponding to LST estimations, a dataset with the associated LST uncertainties, a layer with the data acquisition time, a dataset with the information on the number of processed pixels used on the reprojection and also a quality flag dataset. The latter consists of a 16-bit unsigned integer assigned as described in Table 16 of Annex 2: LST Quality Control Information.

All the LST product datasets are in a common projection and resolution as defined in the user's requirements [AD1 and Chapter 2]: plate-carrée with a resolution of 5/112º.

Details are given into the Product User Manual [CGLOPS1_PUM_LST-V2.2].

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6 RISK OF FAILURE AND MITIGATION MEASURES

The risks associated with CLMS LST product are here classified with respect to effect and frequency following the standards of *The Open Group*[3].

The effects are assessed using the following criteria:

Catastrophic: infers a complete loss of production and no reprocessing capabilities due to missing mandatory input data.

Critical: infers loss of production in a significant part of the globe (e. g. one satellite disk) due to missing mandatory input data with implication on the service performance but no impact on product quality.

Marginal: the product is generated in a degraded mode with possible implications on the product quality but no implications on the service performance.

Negligible: nominal production over the major part of the potential processed areas or a slightly decrease on the product quality.

The frequency could be enumerated as follows:

Frequent: Likely to occur very often.

Likely: Occurs several times over the course of the CGLOPS service.

Occasional: Occurs sporadically.

Seldom: Remotely possible and would probably occur not more than once in the course of the CGLOPS service.

Unlikely: Will probably not occur during the CGLOPS service.

The scheme to assess impact on the LST is defined as:

Extremely High Risk (*E*): The process of global LST will most likely fail with severe consequences.

High Risk (*H*): Significant failure of parts of the LST resulting in certain user requests not being achieved.

Moderate Risk (*M*): Noticeable failure of LST production threatening the success of certain user requests.

[3] http://www.opengroup.org/

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Low Risk (*L*): Certain user requests might not be wholly successful.

These impacts can be derived using the classification schema in Table 10.

Table 10: Risk Classification Schema.

Effect	Frequency				
	Frequent	**Likely**	**Occasional**	**Seldom**	**Unlikely**
Catastrophic	*E*	*E*	*H*	*H*	*M*
Critical	*E*	*H*	*H*	*M*	*L*
Marginal	*H*	*M*	*M*	*L*	*L*
Negligible	*M*	*L*	*L*	*L*	*L*

The main risks associated with the LST production are the failures of HW/SW infrastructure and the missing input data. As described in previous sections (3.2.5.1 and 3.2.6.1) the global LST product requires the following foreign inputs:

- Global Landcover
- LSA-SAF LST (MSG 0° and IODC)
- GEO imagery
- Forecasted atmospheric data
- CLMS FCOVER 300m product
- CLMS Snow cover extent 1 km product

The risk identification and classification as well as the mitigation measures are presented in Table 11.

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Table 11: Risk identification and mitigation assessment.

Risk Context	Risk Identification	Effect	Frequency	Impact	Mitigation Measure
HW/SW infrastructure	HW Failure	Catastrophic	Unlikely	***M***	A mirror system ready to start over the nominal service.
	SW errors	Marginal	Occasional	***M***	Operators and developers ready to identify and correct bugs.
Global Landcover	No	-	-	**-**	Since this is a static database there is no risk of failure thus any mitigation measure needed.
LSA-SAF LST	Missing data	Critical	Occasional	***H***	Nominal LST would be produced but only in the regions not exclusively covered by MSG satellites.
	Missing Service	Critical	Unlikely	***L***	Nominal LST would be produced but only in the regions not exclusively covered by MSG satellites. The process of LST for the MSG satellites in the framework of CGLOPS could be assessed.
GEO Imagery	Missing GOES16 Imagery	Negligible	Occasional	***L***	Use data from backup satellite (GOES14) to produce nominal LST product. LST algorithm prepared to operate with the backup satellite data using a less accurate algorithm.
	Missing GOES-E service	Critical	Seldom	***M***	Use GOES-W imagery to produce nominal LST but the South America region and the East coast of USA would not be processed.
	Missing GOES service	Critical	Unlikely	***L***	Nominal LST would be produced but only in the regions not exclusively covered by GOES satellite.
	Missing Himawari-8 Imagery	Negligible	Occasional	***L***	Use data from backup satellite (Himawari-9) to produce nominal LST product. LST algorithm prepared to operate with the backup satellite data.
	Missing Himawari service	Critical	Unlikely	***L***	Nominal LST would be produced but only in the regions not exclusively covered by Himawari satellite. The use of other satellites e.g. FYI, could be assessed to cover Asia region.
Forecasted atmospheric data	Missing the most recent forecasted data from ECMWF	Negligible	Occasional	***L***	Use the forecasted data from the previous model run with higher step to produce nominal LST.
	Missing ECMWF forecasted data	Marginal	Seldom	***L***	Use ECMWF ERA 5 data to produce LST in degraded mode due to coarser spatial and temporal resolution.
	Missing ECMWF service	Marginal	Unlikely	***L***	Use forecasted data from GFS (NOAA) model to produce LST in degraded mode.
Fraction of Vegetation Cover	Missing the most recent FCOVER RT1 (data with 1 dekad delay)	Marginal	Seldom	***L***	Use a RT6-based (6 dekads delay) FCOVER climatology to account for seasonal vegetation variability
Snow cover extent	Missing the most recent Snow Cover Product	Marginal	Seldom	***L***	Use standard land cover classification

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ANNEX 1: EMISSIVITIES AND THEIR UNCERTAINTIES

Table 12: Emissivities and their uncertainties for the GOES-16 split-window channels, per land cover type. Snow/Ice (land cover 15) values are provided in Table 14 below.

IGBP Land Cover type	TIR1		TIR2	
	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$
1, 2	0.9969 ± 0.0025	0.9685 ± 0.0050	0.9975 ± 0.0025	0.9770 ± 0.0049
3, 4	0.9920 ± 0.0025	0.9685 ± 0.0050	0.9924 ± 0.0025	0.9770 ± 0.0049
5	0.9945 ± 0.0025	0.9685 ± 0.0050	0.9949 ± 0.0025	0.9770 ± 0.0049
6	0.9945 ± 0.0025	0.9687 ± 0.0115	0.9949 ± 0.0025	0.9755 ± 0.0085
7	0.9945 ± 0.0025	0.9687 ± 0.0115	0.9949 ± 0.0025	0.9755 ± 0.0085
8	0.9913 ± 0.0007	0.9687 ± 0.0115	0.9919 ± 0.0004	0.9755 ± 0.0085
9	0.9911 ± 0.0009	0.9687 ± 0.0115	0.9918 ± 0.0006	0.9755 ± 0.0085
10	0.9914 ± 0.0025	0.9914 ± 0.0025	0.9930 ± 0.0028	0.9930 ± 0.0028
11	0.9969 ± 0.0025	0.9685 ± 0.0050	0.9975 ± 0.0025	0.9770 ± 0.0049
12	0.9958 ± 0.0028	0.9731 ± 0.0047	0.9966 ± 0.0028	0.9814 ± 0.0036
13	0.9927 ± 0.0018	0.9581 ± 0.0051	0.9933 ± 0.0016	0.9727 ± 0.0036
14	0.9937 ± 0.0028	0.9731 ± 0.0047	0.9944 ± 0.0028	0.9814 ± 0.0036
16	0.9911 ± 0.0009	0.9551 ± 0.0187	0.9918 ± 0.0006	0.9612 ± 0.0167
17	0.9902 ± 0.0005	0.9902 ± 0.0005	0.9822 ± 0.0002	0.9822 ± 0.0002
18	0.9902 ± 0.0005	0.9902 ± 0.0005	0.9822 ± 0.0002	0.9822 ± 0.0002

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Table 13: Emissivities and their uncertainties for the MSG split-window channels, per land cover type.

IGBP Land Cover type	TIR1		TIR2	
	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$
1, 2	0.9968 ± 0.0022	0.9696 ± 0.0056	0.9973 ± 0.0026	0.9733 ± 0.0076
3, 4	0.9923 ± 0.0022	0.9696 ± 0.0056	0.9922 ± 0.0026	0.9733 ± 0.0076
5	0.9945 ± 0.0022	0.9696 ± 0.0056	0.9947 ± 0.0026	0.9733 ± 0.0076
6	0.9945 ± 0.0022	0.9680 ± 0.0120	0.9947 ± 0.0026	0.9726 ± 0.0092
7	0.9945 ± 0.0022	0.9680 ± 0.0120	0.9947 ± 0.0026	0.9726 ± 0.0092
8	0.9914 ± 0.0009	0.9680 ± 0.0120	0.9917 ± 0.0005	0.9726 ± 0.0092
9	0.9911 ± 0.0013	0.9680 ± 0.0120	0.9915 ± 0.0007	0.9726 ± 0.0092
10	0.9885 ± 0.0055	0.9885 ± 0.0055	0.9924 ± 0.0024	0.9924 ± 0.0024
11	0.9968 ± 0.0022	0.9696 ± 0.0056	0.9973 ± 0.0026	0.9733 ± 0.0076
12	0.9949 ± 0.0026	0.9728 ± 0.0052	0.9967 ± 0.0029	0.9780 ± 0.0031
13	0.9926 ± 0.0019	0.9574 ± 0.0066	0.9930 ± 0.0017	0.9711 ± 0.0037
14	0.9934 ± 0.0026	0.9728 ± 0.0052	0.9942 ± 0.0029	0.9780 ± 0.0031
15	0.9888 ± 0.0068	0.9888 ± 0.0068	0.9649 ± 0.0248	0.9649 ± 0.0248
16	0.9911 ± 0.0013	0.9469 ± 0.0211	0.9915 ± 0.0007	0.9658 ± 0.0068
17	0.9900 ± 0.0006	0.9900 ± 0.0006	0.9856 ± 0.0003	0.9856 ± 0.0003
18	0.9900 ± 0.0006	0.9900 ± 0.0006	0.9856 ± 0.0003	0.9856 ± 0.0003

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Table 14: Emissivities and their uncertainties for the Himawari-8 split-window channels, per land cover type. Snow/Ice (land cover 15) values are provided in Table 14 below.

IGBP Land Cover type	TIR1		TIR2	
	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$
1, 2	0.9969 ± 0.0025	0.9683 ± 0.0055	0.9975 ± 0.0025	0.9772 ± 0.0052
3, 4	0.9920 ± 0.0025	0.9683 ± 0.0055	0.9925 ± 0.0025	0.9772 ± 0.0052
5	0.9945 ± 0.0025	0.9683 ± 0.0055	0.9950 ± 0.0025	0.9772 ± 0.0052
6	0.9945 ± 0.0025	0.9685 ± 0.0114	0.9950 ± 0.0025	0.9756 ± 0.0088
7	0.9945 ± 0.0025	0.9685 ± 0.0114	0.9950 ± 0.0025	0.9756 ± 0.0088
8	0.9914 ± 0.0006	0.9685 ± 0.0114	0.9920 ± 0.0005	0.9756 ± 0.0088
9	0.9911 ± 0.0008	0.9685 ± 0.0114	0.9919 ± 0.0006	0.9756 ± 0.0088
10	0.9916 ± 0.0024	0.9916 ± 0.0024	0.9930 ± 0.0028	0.9930 ± 0.0028
11	0.9969 ± 0.0025	0.9683 ± 0.0055	0.9975 ± 0.0025	0.9772 ± 0.0052
12	0.9958 ± 0.0028	0.9729 ± 0.0048	0.9966 ± 0.0027	0.9814 ± 0.0040
13	0.9927 ± 0.0018	0.9573 ± 0.0051	0.9934 ± 0.0016	0.9721 ± 0.0039
14	0.9937 ± 0.0028	0.9729 ± 0.0048	0.9944 ± 0.0027	0.9814 ± 0.0040
16	0.9911 ± 0.0008	0.9553 ± 0.0188	0.9919 ± 0.0006	0.9588 ± 0.0238
17	0.9901 ± 0.0005	0.9901 ± 0.0005	0.9813 ± 0.0002	0.9813 ± 0.0002
18	0.9901 ± 0.0005	0.9901 ± 0.0005	0.9813 ± 0.0002	0.9813 ± 0.0002

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Table 15: Snow/Ice channel emissivities as a function of viewing angle, for GOES-16 and Himawari.

	GOES-16		Himawari	
SZA (°)	ϵ_{IR1}	ϵ_{IR2}	ϵ_{IR1}	ϵ_{IR2}
0	0.9838 ± 0.0094	0.9687 ± 0.0202	0.9831 ± 0.0099	0.9679 ± 0.0209
10	0.9840 ± 0.0093	0.9690 ± 0.0199	0.9833 ± 0.0098	0.9683 ± 0.0206
20	0.9840 ± 0.0093	0.9689 ± 0.0200	0.9832 ± 0.0098	0.9682 ± 0.0207
30	0.9830 ± 0.0099	0.9671 ± 0.0211	0.9822 ± 0.0104	0.9664 ± 0.0219
40	0.9796 ± 0.0118	0.9606 ± 0.0254	0.9787 ± 0.0125	0.9597 ± 0.0263
50	0.9710 ± 0.0169	0.9438 ± 0.0361	0.9696 ± 0.0177	0.9425 ± 0.0374
60	0.9542 ± 0.0266	0.9113 ± 0.0570	0.9521 ± 0.0280	0.9093 ± 0.0591

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ANNEX 2: LST QUALITY CONTROL INFORMATION

Table 16: Bit-assignment in the LST quality flag.

Decimal value	Sat	Description
0	MULTI-MISSION	Sea pixel
8		Cloud free pixel (clear sky > 90%)
16		Pixel contaminated by clouds (90% > clear sky > 10%)
24		Pixel filled by clouds (clear sky < 10%)
32		Unprocessed pixel
1	GOES	Sea pixel
9		Cloud free pixel (clear sky > 90%)
17		Pixel contaminated by clouds (90% > clear sky > 10%)
25		Pixel filled by clouds (clear sky < 10%)
33		Unprocessed pixel
2	MSG (0° service)	Sea pixel
10		Cloud free pixel (clear sky > 90%)
18		Pixel contaminated by clouds (90% > clear sky > 10%)
26		Pixel filled by clouds (clear sky < 10%)
34		Unprocessed pixel
3	MTSAT or Himawari	Sea pixel
11		Cloud free pixel (clear sky > 90%)
19		Pixel contaminated by clouds (90% > clear sky > 10%)
27		Pixel filled by clouds (clear sky < 10%)
35		Unprocessed pixel
4	MSG IODC service	Sea pixel
12		Cloud free pixel (clear sky > 90%)
20		Pixel contaminated by clouds (90% > clear sky > 10%)
28		Pixel filled by clouds (clear sky < 10%)
36		Unprocessed pixel
64		Pixel out of Disk

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ANNEX 3: CLOUD MASK

This section describes the algorithms used to discriminate clear and cloudy pixels for imagers onboard GOES and Himawari since, for the MSG missions, the cloud masking is performed within LSA-SAF using the SAF-NWC operational algorithm.

Cloud identification is an essential first step in the LST processing chain. The algorithm is based on thresholding techniques applied to visible and infrared channels available in these sensors (Derrien and Le Gléau, 2005). The set of tests correspond to a subset of those used for the Spinning Enhanced Visible and Infrared Imager (SEVIRI) channels. Most thresholds used by the algorithm change between time slots and are obtained from statistics performed over the previous 30-days for the same pixel and observation time. Unlike other methodologies that obtain those thresholds from simulations with fast radiative transfer models using input from Numeric Weather Prediction (NWP) models, we require a minimum of a priori information on the atmosphere and surface conditions.

The algorithm may be summarized in the following steps:

Step 1: Calculate medians of brightness temperatures/reflectance over a 30-days sliding window to be used as an estimation of the clear sky values for a specific timeslot and location;

Step 2: Gather input data for a specific timeslot in the same resolution and projection as the satellite imagery;

Step 3: For each land pixel, compare brightness temperatures and reflectances (and their combined information) to thresholds delimiting clear and cloudy situations. The same thresholds used by the SAF-NWC algorithm for SEVIRI are used here. Depending on the exceedance or not of these thresholds, the pixel is classified as clear or cloudy;

Step 4: A set of spatial tests are applied to the obtained cloud mask to detect small broken clouds or cloud edges, filter spurious isolated cloudy pixels and to filter spurious isolated clear pixels;

Step 5: The final cloud mask is stored in a temporary HDF5 file.

The choice of these thresholds takes into account a few physical assumptions. Clouds generally appear as cold pixels for bands centered around 11 μm, and bright reflectances at 0.6 μm. However, an efficient cloud identification algorithm requires a deeper analysis of several factors, namely: (i) multichannel response to the various types of clouds; (ii) the spatial patterns of TOA measurements and their matching to the texture of the surface / clouds; (iii) and even the evolution of TOA measurements in time. However, there are a few situations, which limit the performance of these algorithms. Identification of low clouds, often with relatively warm tops, during night-time makes use of their relatively low emissivity at the middle infrared bands (MIR; around 3.8 μm). Moreover, the discrimination between snow-covered clear sky pixels and clouds takes into account the low/high reflectivity of snow at the MIR/VIS wavelengths.

The algorithm used in the current version of the LST product is an adaptation of the package used for the sensors used by previous versions. Therefore, it is a simpler version when compared to the full

algorithm used by the SAF-NWC package. It uses three channels, all of which, disseminated via EUMETSAT, as detailed in Table 16.

Table 17: Spectral bands used in the current cloud masking algorithm

	GOES	Himawari
VIS	0.59 – 0.69 µm	0.63 – 0.66 µm
MIR	3.80 – 4.00 µm	3.74 – 3.96 µm
TIR	10.8 – 11.6 µm	11.1 – 11.3 µm

The abovementioned thresholds depend on solar illumination (there are separate values for night-time, twilight and daytime; which correspond to Solar Elevation Angles < -3°, between -3° and 10° and > 10°, respectively), SZA and land cover type. Some of the tests make use of ancillary data, which are retrieved from ECWMF operational forecasts. These are briefly described in Table 17.

Table 18: Tests applied to each pixel to determine if it is cloudy.

Test number	Condition	Notes
Cloud Identification Tests		
1.1 (TIR)	• $T_{TIR} < T_{TIR-Clear} - \Delta$	• $T_{TIR-Clear}$ is estimated using ECMWF forecasted values of skin temperature (SKT), converted to brightness temperatures using a linear regression of the form $T_{TIR-Clear} = A + B.SKT$ (coefficients A and B are calibrated using a similar strategy to the one described in section 3.2.5.2). • $\Delta = 7$ K (night); $\Delta = 5$ K (day); $\Delta = 6$ K (twilight)
1.2 (MIR)	• $T_{MIR} < T_{MIR-Clear} - \Delta$	• $T_{MIR-Clear}$ is estimated as the median of measured T_{MIR} over the previous 30 days, for the same pixel and observation time • $\Delta = 10$ K
1.3 (Cirrus at night)	• Night time • $T_{MIR} - T_{TIR} > 2$ K • $T_{TIR} < 303.15$ K	

1.4 (low clouds shadowed by other clouds)	• $T_{TIR} - T_{MIR} > \Delta$ • $T_{TIR} > 240$ K	• $\Delta = 8$ K over bare soil, $\Delta = 5$ K elsewhere
1.5 (day time)	• Day time • $R_{VIS} > R_{VIS-Clear} - \Delta$	• $R_{VIS-Clear}$ is estimated as the median of measured R_{VIS} over the previous 30 days, for the same pixel and observation time • $\Delta = 0.08$
1.6 (low clouds)	• Day time / twilight • $T_{MIR} - T_{TIR} > T_{MIR-Clear}$ • $T_{TIR} > 240$ K	• $T_{MIR-Clear} = 4.0 + 0.4\,\alpha_{clim}\cos(\theta_{Sun}) + f_{correction}$ • $f_{correction} = 36\cos(\theta_{Sun})\,(\cos(\theta_{scat}) - 0.41)^2$ • θ_{Scat} = sun scattering angle • α_{clim} = climatological surface albedo for the pixel
Spatial Texture Tests		
2.1 (small broken clouds and cloud edges)	• $stdev(T_{TIR}) > \Delta$ • $stdev(T_{TIR} - T_{MIR}) > \Delta$	• $\Delta = 2$ K for day time; $\Delta = 1$ K for night • no mountain regions
2.2 (same as 2.1 but using visible reflectances)	• Day • $Max\left(R_{VIS_i} - R_{VIS_j}\right) > 0.02$	• The difference between R_{VIS} for pixel i is compared to the value of its 8 land neighbours (j) • no mountain regions
Spatial Filtering Tests		
3.1 (isolated cloudy pixels)	• Isolated pixels classified as cloudy by a test using T_{MIR} (i.e. all neighbours are clear)	• Pixel is re-classified as clear
3.2 (isolated clear pixels)	• Isolated pixels classified as cloudy (i.e. all neighbours are cloudy)	• Pixel is reclassified as cloudy

Quality Assessment

The algorithm verification was based on two steps. First, it was applied to SEVIRI data and the output compared with the cloud mask which was obtained with the original full set of tests of the NWC SAF

software. Then, the cloud mask obtained for GOES was compared with that obtained for SEVIRI using the original NWC SAF algorithm, for the overlapping area.

Figure 23 and Figure 24 show the cloud-product applied to the MSG data and the reference NWC SAF Cloud Mask product for a day-time and a night-time case, respectively. This example shows that all cloud features are present in both cloud masks. Similar results were obtained for other timeslots/days. When applied to GOES data (Figure 25 and Figure 26), it is also possible to check that all the main cloud features identified by MSG using the NWC SAF software over South America, are also identified by the adapted cloud mask scheme used here.

The quality of the final LST product may be affected by the performance of the cloud mask scheme. When a pixel is erroneously classified as cloudy, the impact on the final product is only related to a sampling decrease. When a cloudy pixel is erroneously classified as clear, it may severely impact the final product, since in that case we are measuring the cloud top temperature or a mix between cloud top temperature and land surface temperature (in the case of partly cloudy pixels). The error magnitude depends on cloud top height, since higher cloud tops have colder temperatures. Higher clouds are usually easily distinguishable by the IR tests, and frequently in the visible channels as well as they are brighter than the background. Partly cloudy or pixels with low clouds are more prone to misclassification errors, and may lead to LST errors over 5 K.

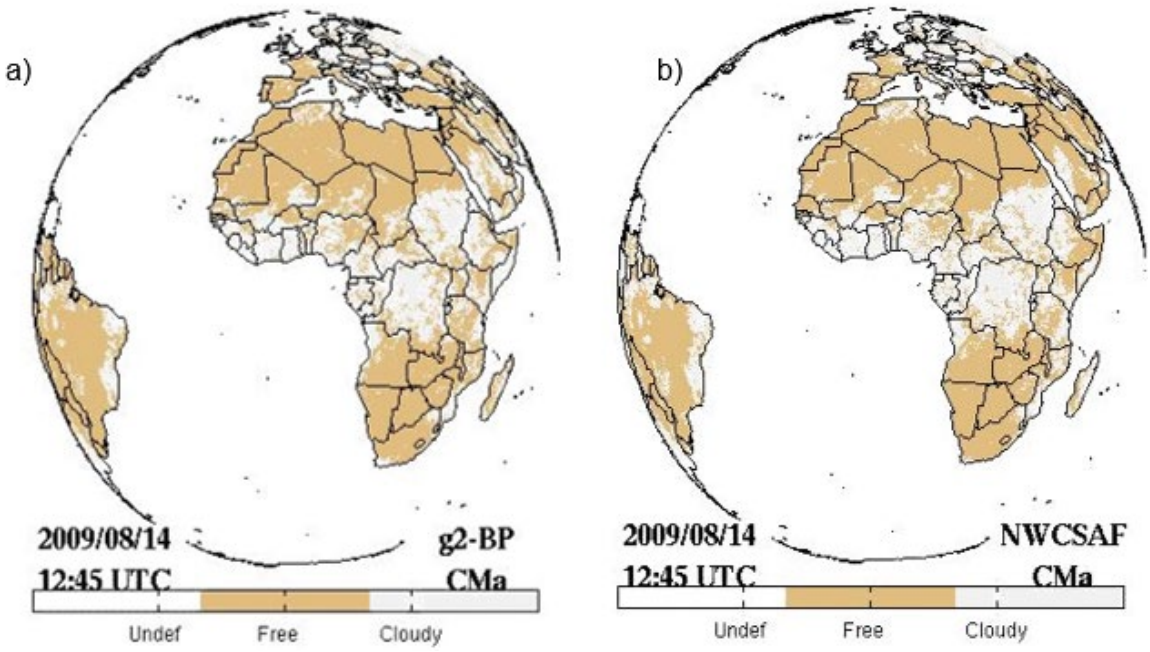


Figure 23 - (a) Cloud-product applied to the MSG data and (b) the NWC SAF CMa product, both for the 14th of August 2009 at 12:45 UTC.

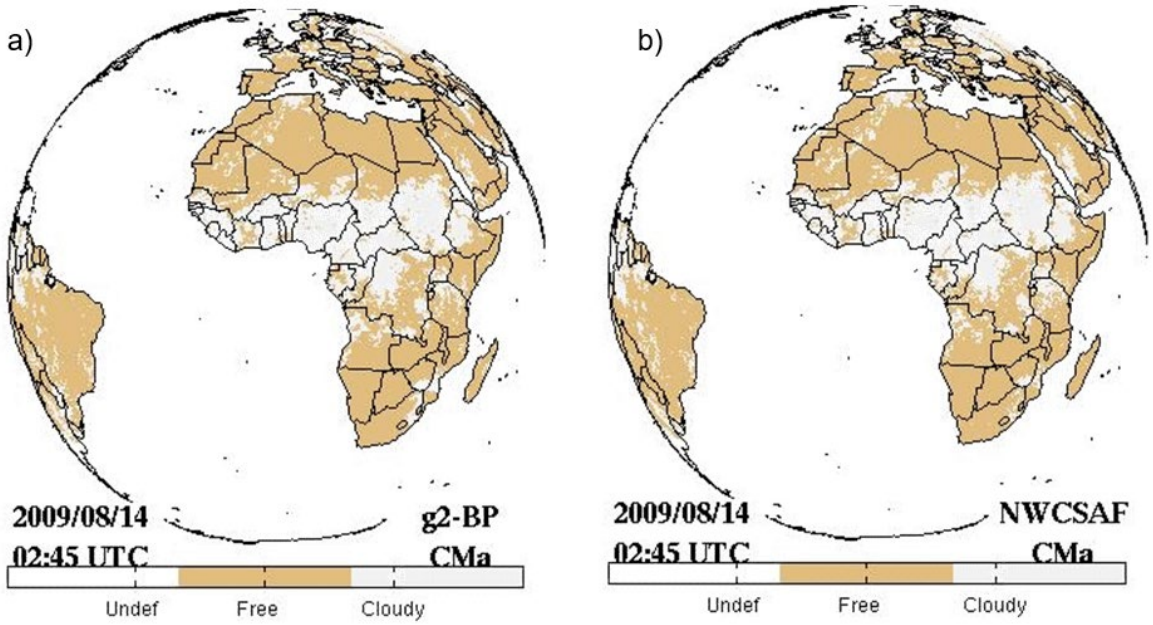


Figure 24 – Same as Figure 23 but at 02:45 UTC.

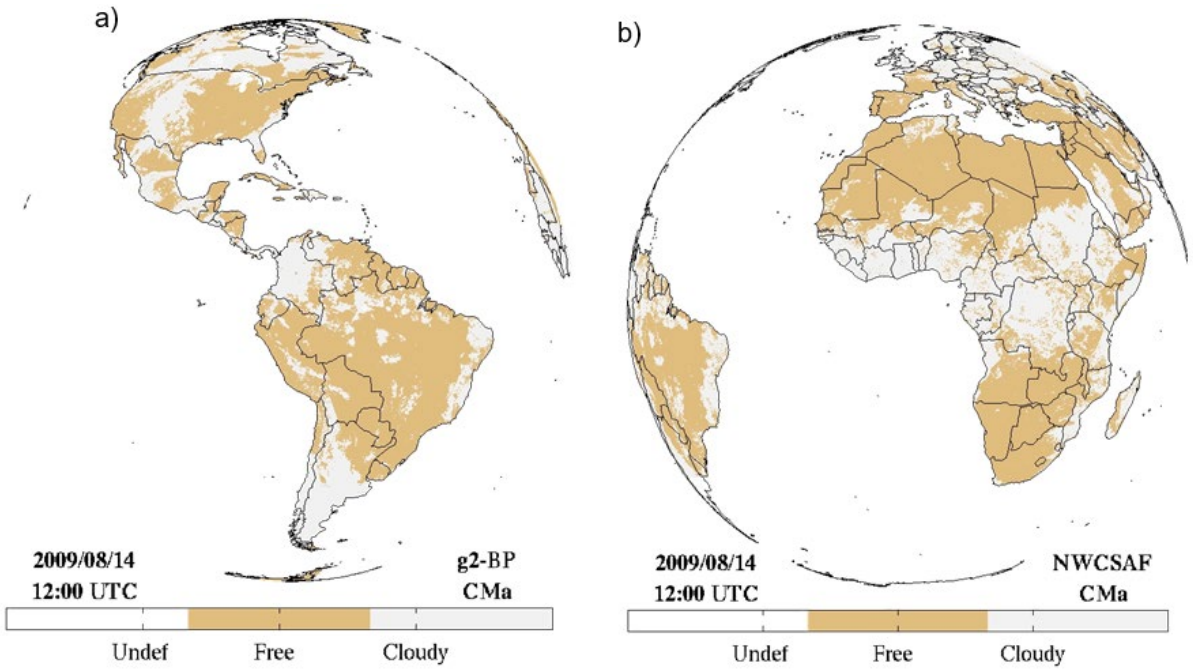


Figure 25 – Cloud product applied to the GOES data (a) and the NWC SAF Cloud Mask product for MSG (b) for the 14th of August 2009 at 12:00 UTC.

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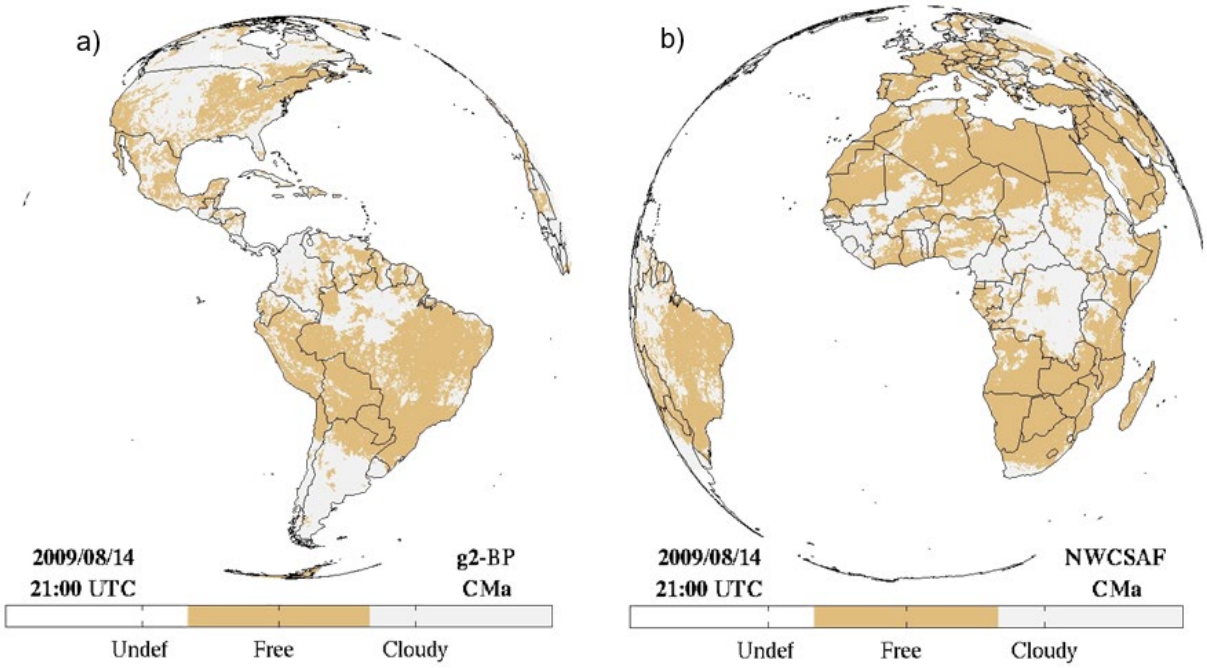


Figure 26 – Same as Figure 25 but at 21:00 UTC.

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ANNEX 4: MIGRATION TO CLMS FCOVER 300M

From version 2.1 onwards, the LST processing chain ingests the CLMS FCOVER 300m product, based on Sentinel-3/OLCI data [CGLOPS1_ATBD_FCOVER300m-V1.1]. As in the case of the 1 km product used in version 2.0, there are some differences between the LSA-SAF FVC product and the CLMS counterpart. Data from the first dekad of January and the first dekad in July were used to provide an estimate of the differences between the two products, which are illustrated in Figure 27. In January, there is an underestimation of the former over Amazonia (likely due to high cloud cover and atmospheric water vapor content, combined with high viewing angles), and an underestimation over the East and South East Brazil. In July, the differences over the central part of the domain are alleviated, while over the Inter-Tropical Convergence Zone (ITCZ) some negative (positive) differences are observed for north South America (West Africa). As demonstrated in the QAR [CGLOPS1_QAR_LST-LST10-V2.1], differences within this range have a limited effect in the final LST retrieval, and can be considered as part of the uncertainty due to emissivity.

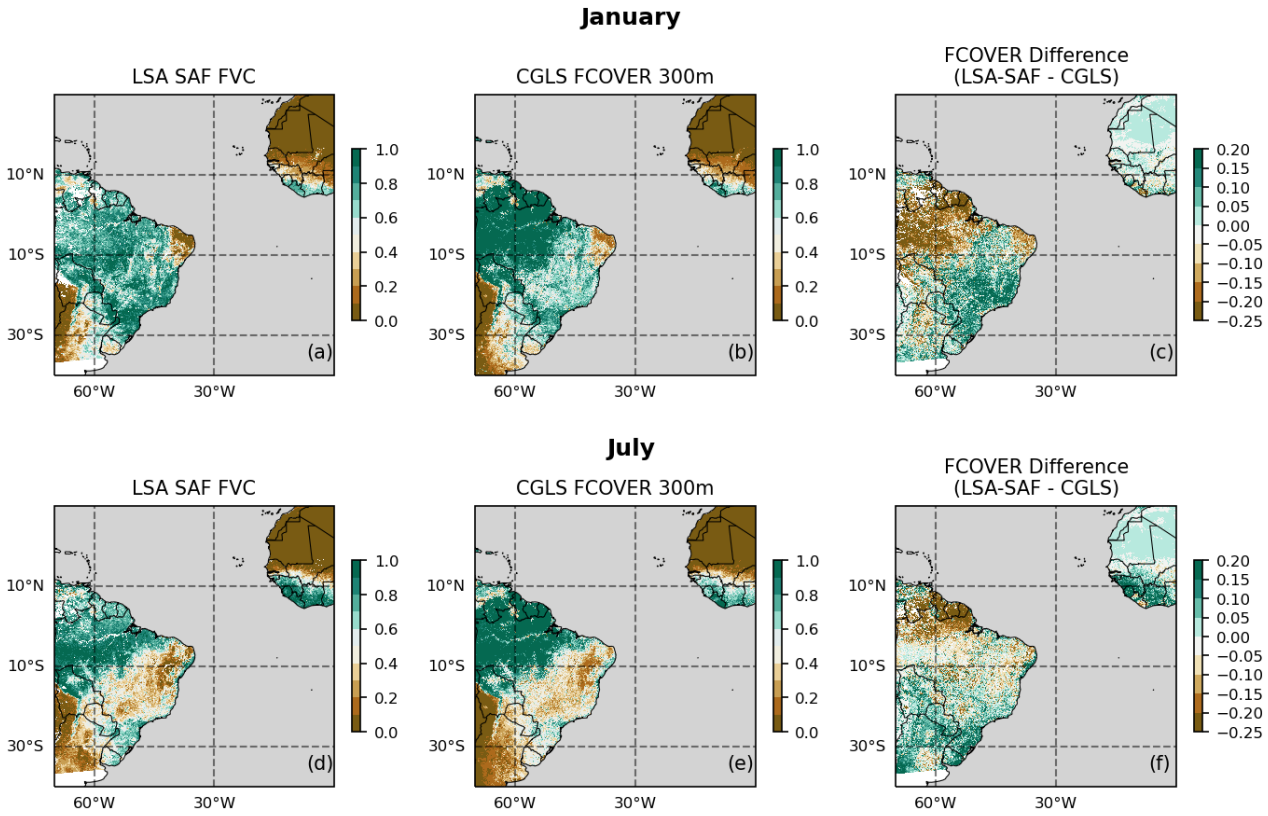


Figure 27 – Comparison of Fraction of Green Vegetation Products from LSA SAF (left) and CLMS 300m (center). Their difference is shown in the right column. In the top row, average data from 1-10 Jan 2021 for LSA SAF FVC and 10 Jan 2021 dekad for CLMS FCOVER were used, while in the bottom row, average data from 1-10 July 2021 for LSA SAF FVC and 10 July 2021 dekad for CLMS FCOVER were used.

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